Research Article Vol. 13, No. 3 / March 2026 / Optica 491

OPTICA

Dynamic motion-blur suppression and super-resolution localization of UAV targets using single-photon LiDAR under ultra-low SNR

Dongfang Guo,[1] Xin Zhou,[1,2,3] Jianfeng Sun,[1,2,*] Sheng Chen,[1,2] Yinbo Zhang,[1] Sining Li,[1] and Yanchen Qu[1]

[1]National Key Laboratory of Laser Spatial Information, Harbin Institute of Technology, Harbin, Heilongjiang 150001, China
[2]Zhengzhou Research Institute of Harbin Institute of Technology, Zhengzhou, Henan 450000, China
[3]Research Center for Space Optical Engineering, Harbin Institute of Technology, Harbin, Heilongjiang 150001, China
*sjf@hit.edu.cn

Received 9 October 2025; revised 6 January 2026; accepted 1 February 2026; published 10 March 2026

Long-range detection and localization of UAVs remain extremely challenging, as motion-induced spatiotemporal distortion and weak echo returns severely degrade imaging quality. Although single-photon LiDAR offers photon-level sensitivity and long-range capability, its performance in dynamic UAV detection under low SNR remains constrained by motion blur, sparse photon measurements, and detector resolution, leading to unreliable target detection and localization. We present a five-dimensional motion-aware reconstruction (5D-MR) framework that jointly models the temporal (time-of-flight), spatial (row and column position), photon-count, and frame-sequence features of triggered photon events to perform signal preprocessing and dynamic target detection. Signal photons are robustly extracted from an overwhelming background via a two-stage RANSAC temporal fitting followed by DBSCAN clustering in the frame domain. Trajectory prediction combined with subpixel interpolation yields motion compensation and super-resolved localization beyond the sensor's native resolution. Simulation results demonstrate that, even at an SNR of −9.03 dB with 0.5343 photons per pixel, dynamic targets are localized with angular errors below 0.1 pixel and range errors within 0.5 m. Field experiments further validate the method for a UAV target at 1.8 km, achieving motion-blur suppression and super-resolution location reconstruction at an SNR of −8.97 dB with only 0.3550 photons per pixel, resulting in angular precision better than 0.01° and size estimation errors below 0.02 m, representing nearly two orders of magnitude improvement in resolvable localization accuracy. The 5D-MR framework enables accurate 3D localization of dynamic UAVs under extreme photon-starved conditions, surpassing detector-array resolution limits and extending single-photon LiDAR to remote airspace monitoring and ultra-low-SNR dynamic imaging. © 2026 Optica Publishing Group under the terms of the Optica Open Access Publishing Agreement

https://doi.org/10.1364/OPTICA.581370

1. INTRODUCTION

The rapid proliferation of low, small, and slow UAVs has dramatically increased the number of airborne targets, which are characterized by small radar cross sections (RCSs, typically −20 to −10 dBsm, corresponding to 10^{-3}–10^{-1} m^2), low flight altitudes, and high maneuverability, significantly complicating low-altitude surveillance [1–3]. Conventional long-range sensing systems face limitations in detection range or localization accuracy under conditions of long distance, low signal-to-noise ratio (SNR), low elevation angles, and complex environments. For consumer-grade UAVs, commercial short- and medium-range radars typically detect targets within 0.5–3 km [4], while high-power, phased-array, or large-aperture systems can extend the range. However, detection probability drops sharply when the target RCS approaches −20 dBsm or single-pulse SNR falls below 5 dB, while miss rates increase further under SNR < 0 dB with insufficient pulse accumulation [5,6]. RF monitoring relies on UAV control or data-link signals, typically achieving detection ranges of 0.1–3 km [7]. Under high SNR and multi-antenna/multi-station setups, azimuthal precision can reach about 1°, but errors often increase to several degrees under urban multipath or low SNR conditions. Detection becomes nearly impossible if the UAV disables transmission or employs low-duty-cycle/frequency-hopping links [8]. Passive optical imaging range can extend to 2 km in high-contrast daytime scenes but drops to tens to hundreds of meters under haze or nighttime conditions [9,10]. Consequently, achieving reliable detection and high-precision localization of UAVs at long range under low-SNR conditions remains a critical challenge for airspace surveillance.

Single-photon detection technologies offer a promising solution for long-range detection of such targets [11–13]. Geiger-mode avalanche photodiode (Gm-APD)-based LiDAR provides photon-level sensitivity [14] and enables 3D scene reconstruction through

point, line, or flash scanning [15,16]. Previous studies demonstrated static imaging of extended targets at distances up to 201.5 km with an average of 0.44 photons per pixel, leveraging long-time acquisition and high-frame accumulation, highlighting its ultra-long-range sensing potential [17]. With the expansion of Gm-APD arrays, large field-of-View (FoV) 3D acquisition can be achieved without mechanical scanning, enhancing dynamic response capability [18–20]. Nevertheless, despite the advantages of SPL, the limited size of detector arrays still constrains spatial resolution, impeding accurate reconstruction of small targets such as UAVs and degrading their observable imaging features [21,22]. Under dynamic detection, target motion during the sampling period leads to the energy of the return photons spreading in space, and the projection on the detector plane causes serious X–Y motion blur. This effect reduces localization accuracy and geometric reconstruction quality, adversely affecting subsequent recognition and trajectory estimation. At long range and low SNR, photons become extremely sparse and easily buried in background noise, further complicating dynamic target detection and localization.

Existing approaches address these challenges along three main directions. First, sampling-rate enhancement, such as MHz-level laser repetition by McCarthy *et al.* or pseudo-random coding for fast ranging proposed by Zhang *et al.* [23], shortens acquisition time and improves dynamic response but is limited by transmission loss at long range [24]. Second, image-domain reconstruction enhances target recovery via filtering, Bayesian modeling [25], or gating strategies [26], improving data utilization but often neglecting the Poisson statistics of photon counts. Third, spatiotemporal trajectory modeling, e.g., 3D photon-event tensors [27] or Hough-transform-based motion extraction [28], enables motion-blur suppression and dynamic detection. Despite progress, these methods struggle to extract targets and achieve high-precision localization under extremely low-SNR conditions (e.g., below 0 dB [29]), mainly because they rely on 2D image-domain processing or hardware stacking without jointly modeling photon statistics across time, space, and frame sequence. Temporal consistency across frames and inter-pixel trajectory continuity are often ignored, impeding robust signal-background separation and the utilization of dynamic inter-frame information.

To overcome the dual limitations of motion blur and low SNR in long-range dynamic UAV detection, we propose a five-dimensional joint-domain dynamic reconstruction framework, termed five-dimensional motion-aware reconstruction (5D-MR). Within local sliding windows, photon events are jointly modeled across temporal, spatial, photon-count, and frame-sequence features, with inter-frame and inter-pixel spatiotemporal consistency constraints. This enables robust photon extraction, trajectory prediction, and subpixel reconstruction. Monte Carlo simulations show that, even at an SNR as low as −9.03 dB, our method successfully recovers the true positions of dynamic point targets moving at 4 m/s (cross/column direction) and 7 m/s (depth direction) at 1 km, with angular errors below 0.1 pixel and ranging errors under 0.5 m. Field experiments further validate motion-blur suppression for a UAV at 1.8 km under extreme conditions (SNR = −8.97 dB, 0.3550 photons per pixel), achieving super-resolution localization errors below 0.01° and size estimation errors below 0.02 m. Compared with existing reconstitution methods, our approach extends the usable SNR threshold by nearly 9 dB.

This work, for the first time, overcomes the critical challenges of motion-induced blur and intrinsic spatial resolution limits in SPL for UAV detection, achieving accurate localization of UAV targets under extreme photon-starved, long-range conditions. By addressing these fundamental limitations, SPL is rendered practically viable for real-time airspace monitoring and high-fidelity dynamic target surveillance, paving the way for photon-efficient long-range 3D sensing of small and agile aerial platforms.

2. DYNAMIC DETECTION MODELING OF Gm-APD SPL

The detection process of the Gm-APD SPL is based on high-frequency triggered photon events and their statistical modeling, as illustrated in Fig. 1. SPL employs a time-to-digital converter (TDC), which is integrated within the signal processing and control module, to record the time of flight (ToF) of triggered photon events [30]. By adjusting the gating delay and width, the system ensures sufficient depth coverage. Subsequently, time-correlated single photon counting (TCSPC) statistics are constructed by accumulating time-stamped events over multiple repetitions, forming histograms from which target intensity and range information are extracted.

Due to the limited FoV of SPL, the imaging size of a long-range UAV is often smaller than a single pixel, effectively appearing as a point or sub-pixel target [31]. The time-dependent LiDAR equation for the echo power can be expressed as [32]

$$P_r(t) = \frac{4P_t(t-t_d)}{\pi\theta_B^2 R^2}\rho A_T \frac{A_r}{\pi R^2}\eta_{\text{atm}}\eta_{\text{sys}}, \tag{1}$$

where P_t is the transmitted laser power, ρ is the target reflectivity, θ_B is the beam divergence, R is the range to the target, t_d is the round-trip time, A_t is the effective target area, A_r is the effective receiver aperture area, η_{atm} is the two-way atmospheric transmittance, and η_{sys} is the overall system efficiency, including optics and detector losses.

Assuming the laser pulse width is p_w, and the transmitted pulse energy is uniformly distributed without distortion, the mean number of detected signal photons is

$$N_s = \frac{\eta_q \int_{t_d}^{t_d+p_w} P_r(\tau|i)\mathrm{d}\tau}{h\upsilon}, \tag{2}$$

where η_q is the internal quantum efficiency, i is the index of the laser pulse, $h\upsilon$ is the single-photon energy, and $P_r(\tau|i)$ denotes the conditional received optical power density as a function of time τ under the i-th transmitted pulse.

The background noise is composed of ambient radiation and detector dark counts. Ignoring dark counts, the mean number of background photons is

$$N_n = \frac{\eta_q \int_0^{\Delta t} \eta_r \Omega_R I_{bk}\mathrm{d}\tau}{h\upsilon}. \tag{3}$$

The SNR is defined as [32]

$$\text{SNR} = 10\lg\left(\frac{N_s}{N_n}\right). \tag{4}$$

As a triggered detector, the Gm-APD outputs a saturated avalanche pulse whenever one or more photons are absorbed, thereby increasing the photon count by one [33]. The probability of a pixel being triggered is given by [32]

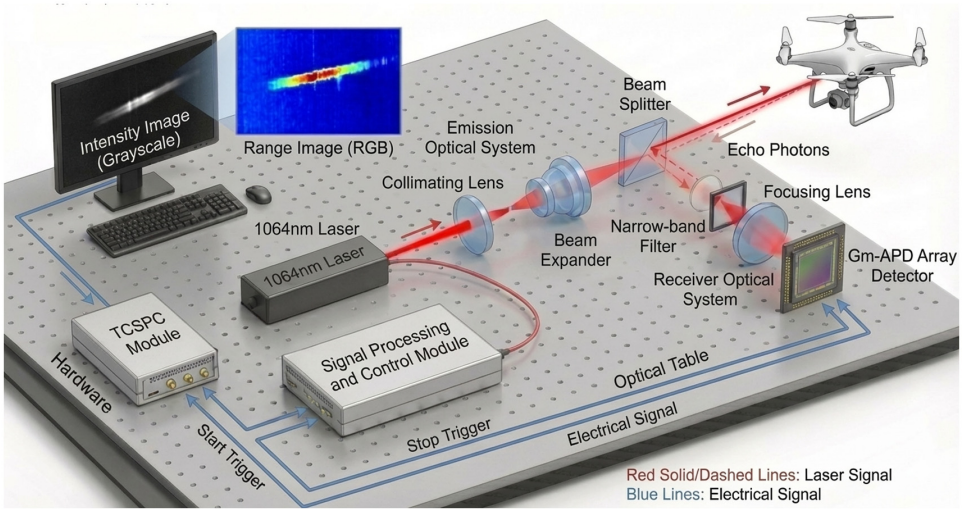


Fig. 1. Detection principle of Gm-APD SPL.

$$p_{\text{elec}}(h) = 1 - p_{\text{elec}}(0) = 1 - \exp\left(-N_n\right)\left(\langle N_s\rangle / M' + 1\right)^{-M'}, \tag{5}$$

where the parameter M' is a statistical shape/dispersion parameter used in the triggering probability model for the Gm-APD detector. It accounts for the over-dispersion of photon arrival statistics (e.g., speckle and intensity fluctuations) relative to a pure Poisson model.

With a TDC precision of Δt, the gating window is divided into $J = T_g/\Delta t$ bins, where T_g is the gate width. The probability that a pixel is triggered at the hth bin during the ith laser pulse is

$$\begin{aligned} p_{h|i} &= \left[\prod_{l=0}^{h-1} p_{\text{elec}}(0)\right]\left(1 - p_{\text{elec}}(0)\right) \\ &= \left[\prod_{l=0}^{h-1} \exp(-N_{n,l})\left(N_{s,l}/M' + 1\right)^{-M'}\right] \\ &\quad \times \left[1 - \exp(N_{n,h})\left(N_{s,h}/M' + 1\right)^{-M'}\right], \quad h = 1, 2 \ldots J. \end{aligned} \tag{6}$$

Therefore, when the sampling period is T_{EX} and the corresponding detection times are floor(T_{EX}/T_D), the average trigger probability is

$$p_h = \sum_{i=1}^{\text{floor}\left(\frac{T_{EX}}{T_D}\right)} p_{h|i} / \text{floor}\left(\frac{T_{EX}}{T_D}\right). \tag{7}$$

For static targets, range, reflectivity, and echo energy remain stable within the sampling cycle, yielding nearly constant triggering probabilities. In contrast, for dynamic targets moving along the depth or across the FoV during sampling, their projected positions on the detector array shift continuously. Such motion alters the spatio-temporal characteristics of laser echoes. On one hand, the target spans multiple pixels during integration, producing cross-pixel spatial blurring; on the other hand, variations in propagation path lengths cause slight shifts in photon arrival times, resulting in temporal broadening or trailing. The echo distribution can thus be written as

$$\begin{cases} N_{s,(x_{i-1},y_{i-1})} = N_s \cdot \omega_{i-1}(t) \\ N_{s,(x_i,y_i)} = N_s \cdot \omega_i(t) \end{cases}, \tag{8}$$

where $\omega_i(t)$ denotes the instantaneous energy distribution, characterizing the scattering ratio into position (x_i, y_i) at t. Each (x_i, y_i) uniquely identifies a spatial position in the observed field.

Because the spatial distribution of SPL echoes approximates a Gaussian profile, we model the cross-pixel diffusion of target energy during motion with a two-dimensional Gaussian function:

$$\omega_i(t) = \frac{1}{2\pi\sigma_x\sigma_y Z(t)} \exp\left(-\left[\frac{(i\Delta x - x_i)^2}{2\sigma_x^2} + \frac{(i\Delta y - y_i)^2}{2\sigma_y^2}\right]\right), \tag{9}$$

where σ represents the diffusion scale, and $Z(t)$ is the normalization factor that ensures $\sum_i \omega_i(t) = 1$. All indices and parameters used in the detection modeling of Gm-APD SPL are listed in Table 1.

Traditional reconstruction methods such as the matched filtering method, peak method, or maximum likelihood estimation method [34,35], rely on frequency histograms accumulated over multiple frames. In dynamic scenarios, however, they can only reconstruct averaged positions or blurred trajectories, failing to recover instantaneous target states.

Fortunately, the nanosecond-level timing resolution and high laser repetition rates of Gm-APD systems allow photon drifts to manifest as continuous temporal variations across neighboring pixels. This provides a foundation for motion analysis. By exploiting multi-frame spatio-temporal registration and motion compensation, high-resolution reconstruction of dynamic targets and effective suppression of motion blur become feasible.

3. PROPOSED METHOD: FIVE-DIMENSIONAL MOTION-AWARE RECONSTRUCTION (5D-MR)

Although Gm-APD SPL can suppress part of the background noise by adjusting gate delays and widths, wide gating is unavoidable in dynamic detection to ensure sufficient depth coverage. This results in significant overlap between target echoes and background noise, substantially reducing the effective SNR. Target motion further causes photon sparsity, temporal drift, and cross-pixel spreading, which exacerbate motion blur and reconstruction distortion.

Table 1. Indices and Parameters Used in the Dynamic Detection Modeling

Index	Parameter / Variable	Index	Parameter / Variable	Index	Parameter / Variable
i	Laser pulse index	h	Timing histogram bin	x, y	Horizontal / vertical position in scene
r, c	Row / column of sensor	f	Frame index	$t_{r,c}^{(i)}$	ToF of ith photon event at pixel (r, c)
$N_{f,r,c}$	Total photon events at (r, c) in frame f	$P_r(t)$	Received optical power	P_t	Transmitted laser power
ρ	Target reflectivity	θ_B	Laser beam divergence	R	Range to target
t_d	Round-trip time	A_t	Effective target area	A_r	Receiver aperture area
η_{atm}	Atmospheric transmittance	η_{sys}	Overall system efficiency	p_w	Laser pulse width
η_q	Quantum efficiency	$h\nu$	Single-photon energy	N_s	Mean number of signal photons
N_n	Mean number of background photons	$p_{\text{elec}}(h)$	Pixel trigger probability	M'	Photon dispersion parameter
Δt	TDC precision	J	Number of bins in gating window	T_g	Gate width
$p_{h\|i}$	Trigger probability at hth bin for ith pulse	T_{EX}	Exposure time	T_D	Laser repetition period
$\omega_i(t)$	Target energy distribution over pixels	σ_x, σ_y	Gaussian diffusion scale	$Z(t)$	Normalization factor

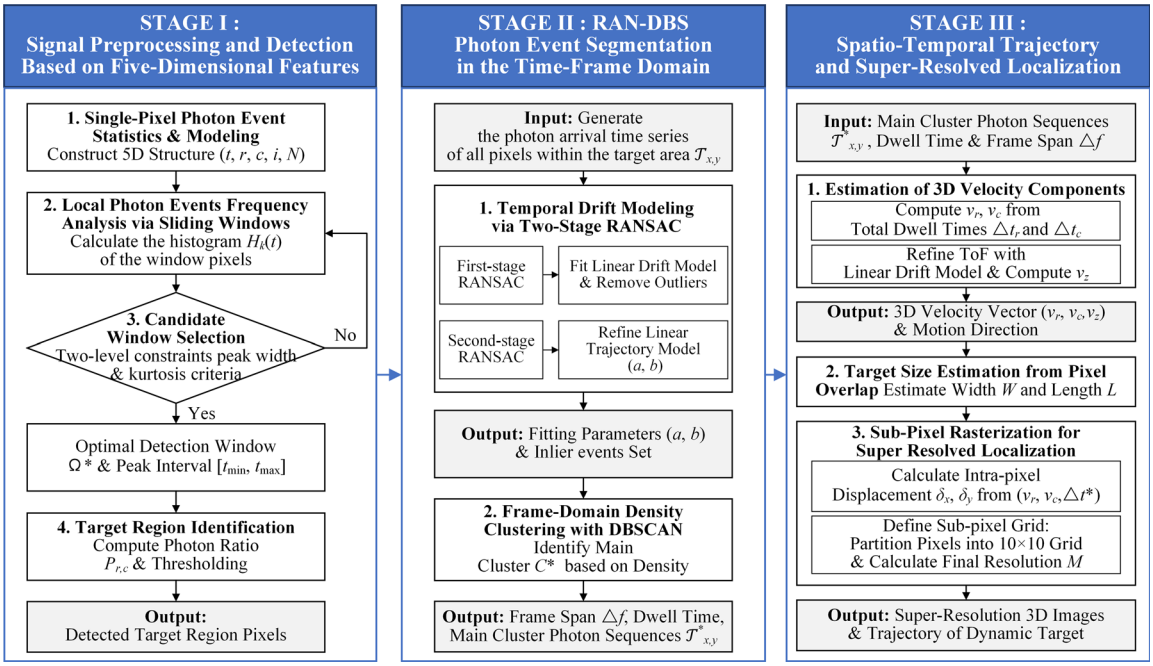


Fig. 2. Processing flow of the 5D-MR framework processing algorithm.

To address the aforementioned challenges, we propose a novel computational imaging framework, termed five-dimensional motion-aware reconstruction (5D-MR), for processing SPL echoes. The 5D-MR framework consists of four sequential stages. First, the echo signals are preprocessed and targets are detected based on joint five-dimensional features in the time, space, frequency, and frame domains. Second, a two-dimensional temporal-frame thresholding and clustering algorithm is applied to separate target photons from noise events and to correct modeling biases. Third, spatiotemporal trajectory estimation across pixels is incorporated to suppress motion blur. Finally, a two-dimensional gridded reconstruction is performed to generate super-resolution localization images of the dynamic target. The overall workflow of the proposed 5D-MR is illustrated in Fig. 2.

A. 5D-MR: Signal Preprocessing and Detection Based on Five-Dimensional Features

1. Single-Pixel Photon Event Statistics and Modeling

In low-SNR conditions, target signals are often buried in global histograms of photon-event counts, as background noise triggers many more pixels than the few target pixels. At the single-pixel level, however, target echoes appear as concentrated peaks in the temporal histogram due to their localized ToF distribution. These peaks serve as the primary feature for detection and statistical modeling. To formalize this process, we denote the photon arrival times recorded at pixel (r, c) in frame f be

$$T_{f,r,c} = \{t_{r,c}^{(i)}\}, \quad i = 1, 2, \ldots, N_{f,r,c}, \tag{10}$$

where $t_{r,c}^{(i)}$ is the ToF of the ith photon event, and $N_{f,r,c}$ is the total number of events recorded at that pixel (r, c) in frame f.

A five-dimensional data structure containing information such as time-space-frequency-frame is constructed by recording the position (r, c) of pixels in the FoV, the arrival time $t_{r,c}^{(i)}$ of photon events in the gate, the trigger index i corresponding to the photon events, and the total number $N_{f,r,c}$ of photon events in any tag in the gate, which is used to describe the temporal and spatial evolution characteristics of dynamic targets.

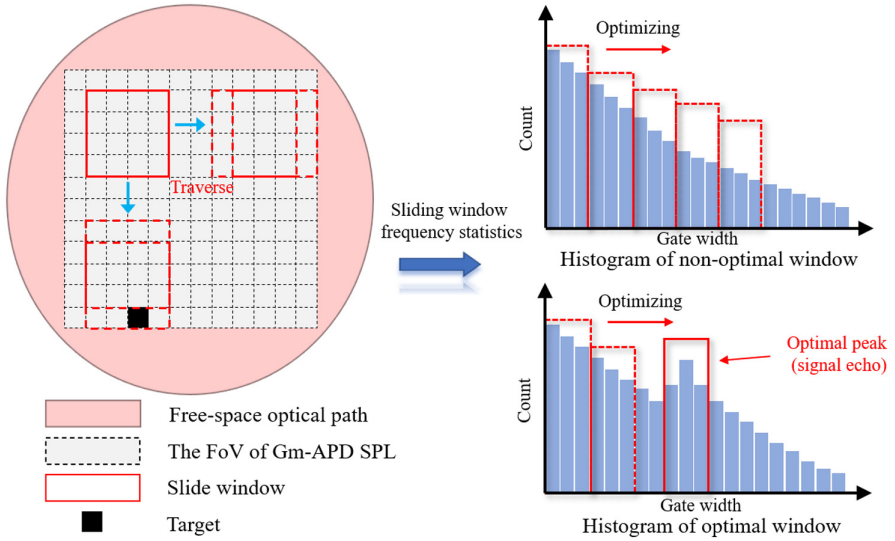


Fig. 3. Illustration of local sliding-window photon statistics for target enhancement.

2. Local Photon Events Analysis via Sliding Windows

To enhance the detectability of sparse targets, a local sliding-window strategy is applied. As shown in Fig. 3, a window of size $W_s \times W_s$ pixels is moved across the FoV with a stride of one pixel. For each sliding window, the frequency distribution histogram of photon events triggered by all pixels in the window is counted:

$$H_k(t) = \sum_{(r,c)\in\Omega_k} \sum_{i=1}^{N_{f,r,c}} \mathbb{I}(t_{r,c}^{(i)} = t), \tag{11}$$

where Ω_k denotes the set of pixels in the kth window and $\mathbb{I}(\cdot)$ is the indicator function.

3. Candidate Window Selection via Peak Statistics

A smoothed probability density curve of photon arrival time is obtained by weighted averaging of the histogram. To select candidate regions, two constraints strategies are imposed:

- Peak-width constraint: considering that typical UAVs fly at speeds below 50 m/s [36], with frame rates of 5–10 Hz, the inter-frame displacement is usually under 10 m. For a gate width of 1000 ns (≈ 150 m depth), the genuine ToF peak should be narrowly concentrated. We therefore set a maximum peak width of 100 bins (≈ 15 m depth) to filter out noise clusters.
- Kurtosis-based criterion: the sharpness of the ToF distribution is quantified by the kurtosis:

$$\mathrm{Kurt}(\mathrm{p_k}) = \frac{1}{\mathrm{N}} \sum_{\mathrm{i}=1}^{\mathrm{N}} \left(\frac{\mathrm{t_i} - \mu}{\sigma} \right)^4, \tag{12}$$

where μ and σ are the mean and standard deviation of the distribution $p_k(t)$, respectively. Higher kurtosis indicates a sharper and more localized peak.

The sub-window with the maximum kurtosis that satisfies the peak-width constraint is chosen as the optimal detection window Ω^*. And the corresponding main peak gating time interval $[t_{\min}, t_{\max}]$ is recorded for subsequent signal photon event extraction and target detection.

4. Target Region Identification via Photon Statistics

For each pixel in the optimal window Ω^*, the ratio $P_{r,c}$ of photon events within the selected peak interval to the total number $N_{r,c}^{\mathrm{peak}}$ of events is computed:

$$\begin{cases} N_{r,c}^{\mathrm{peak}} = \sum_{i=1}^{N_{f,r,c}} \mathbb{I}(t_{r,c}^{(i)} \in [t_{\min}, t_{\max}]), \\ P_{r,c} = \frac{N_{r,c}^{\mathrm{peak}}}{N_{r,c}}. \end{cases} \tag{13}$$

Pixels with $P_{r,c}$ exceeding a threshold γ are classified as target pixels, forming the spatial extent of the detected target region.

B. 5D-MR: RAN-DBS Photon Event Segmentation in the Time–Frame Domain

After detection, the photon arrival time recorded by the pixel in the target area during sampling and its sub-frame index form a two-dimensional sequence $\mathcal{T}_{x,y}$:

$$\mathcal{T}_{x,y} = \left\{ \left(i, t_{x,y}^{(i)} \right) \right\}. \tag{14}$$

Photon events detected by the Gm-APD include both target-reflected signal photons and background noise such as ambient light and detector dark counts. Since the gating width is typically on the order of microseconds, the target signal occupies only a small fraction of the detection window, while background photons are approximately uniformly or Poisson-distributed over time.

Signal-triggered photons exhibit a stable time-of-flight trend across frames: as the target moves in depth or laterally, the arrival-time sequence shows an approximately linear drift or near-constant trend with respect to subframe indices, resulting in a stable linear distribution within the gating window.

Based on these observations, we propose a two-stage photon-event segmentation strategy. A hierarchical combination of two-stage RANSAC (random sample consensus) and DBSCAN (density-based spatial clustering of applications with noise) is employed to model photon densities in both the temporal and frame domains, enabling precise separation of target photons from background noise.

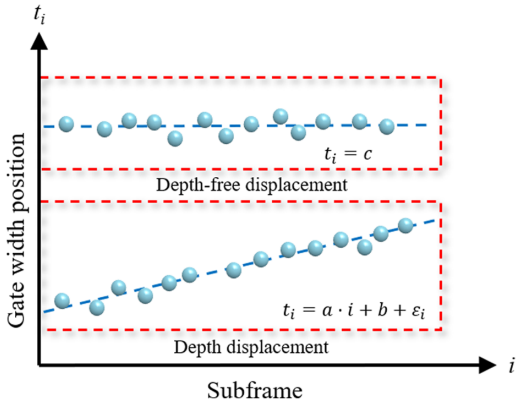


Fig. 4. Schematic diagram of the distribution of photon arrival time within the frame threshold during sampling time.

1. Temporal Drift Modeling via Two-Stage RANSAC

Target motion in depth produces an approximately linear trend in the ToF sequence, as shown in Fig. 4:

$$t_i = a \cdot i + b + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2). \tag{15}$$

Among them, a and b, respectively, represent the time-drift trend (slope and intercept) caused by the longitudinal movement of the target, depicting the deep movement trend; ε_i is the measurement error. It is worth noting that the RANSAC fitting process is guided by the physical characteristics of the SPL. Specifically, the slope parameter a is bounded by the maximum feasible longitudinal velocity of the target, as determined by the time-of-flight relation between photon arrival time and range. The intercept b is bounded within a temporal interval of two laser pulse widths, accounting for the finite temporal extent of the emitted pulse. The residual tolerance ε_i represents the intrinsic timing uncertainty of the detection system and is dominated by electronic jitter and noise. These physically motivated constraints effectively prevent the RANSAC procedure from converging to spurious temporal trends induced by noise photons while preserving robustness under low-SNR conditions.

To robustly extract this trend under low SNR, a two-stage RANSAC procedure is applied:

- **First-stage RANSAC:** applied to the time–frame index sequence to fit a linear drift model of the target's main-photon peak, removing outlier noise points that deviate from the target trend along the temporal axis.
- **Second-stage RANSAC:** further iteratively applied to the inlier set from the first stage to refine the linear time-drift trajectory model of the target, improving fitting accuracy and stability.

The output of the second-stage RANSAC algorithm includes:

- **Linear fitting parameters a and b:** the slope a characterizes the target's depth-wise motion trend during the observation period. According to the time-of-flight principle, photon arrival time is proportional to the target range, while subframe indices correspond to the observation sequence. Therefore, the slope a can be used to estimate the target's average velocity along the laser line-of-sight, with its sign indicating the motion direction.
- **Inlier set:** all photon events consistent with the fitted motion trend, which serve as input for subsequent spatial-domain clustering analysis.

2. Frame-Domain Density Clustering with DBSCAN

Residual noise photons near the ToF trend are further separated using DBSCAN clustering in the frame domain. The neighborhood radius σ is set equal to the effective laser pulse width, such that photon events originating from the same laser pulse fall within a single temporal neighborhood. Together with a minimum point threshold minPts, locally dense photon clusters corresponding to valid target returns are identified:

$$\begin{cases} \text{DBSCAN}(X;\, \sigma,\, \text{minPts}) \to \{C_k\}_{k=1}^{K}, \\ C^* = \underset{C_k}{\text{argmax}} |C_k|. \end{cases} \tag{16}$$

The minimum and maximum frame indices of the main cluster correspond to the frames when the target enters and exits the detection region, producing stable echoes, denoted as f_e and f_s, respectively. Their difference, $\Delta f = f_e - f_s$, reflects the actual dwell time of the target within this region (frame span). The photon arrival-time sequences corresponding to each frame index in the main cluster, $\mathcal{T}^*_{x,y} = \{(i, t^{(i)}_{x,y})\}$, are preserved for subsequent range correction and trajectory modeling.

It is worth emphasizing that no explicit "subframe-index threshold" is applied to remove noise events. Instead, the statistical clustering property in the frame domain is exploited to automatically identify the main cluster of target events in the frame–time joint domain based on a density constraint. This strategy effectively implements a "sparse suppression mechanism" in the frame domain, further distinguishing target signals from coincident but spatially inconsistent background photons, thereby improving the overall robustness and accuracy of the modeling.

C. 5D-MR: Spatio-Temporal Trajectory and Super-Resolved Localization

To address the limitations of low-resolution photon-counting images, the 5D-MR framework integrates photon distributions, sub-frame timing, and dwell statistics across neighboring pixels, and establishes a statistical–geometric model that enables trajectory estimation and sub-pixel localization at the reconstruction endpoint.

1. Estimation of 3D Velocity Components

The velocity components along the row, column, and depth directions are inferred from photon dwell times across multiple pixels. The total dwell times Δt_r and Δt_c are obtained from the frame spans Δf and the imaging frame period T_{frame}:

$$\begin{cases} \Delta t_r = [\Delta f_{t_1} + \Delta f_{t_2} + \cdots + \Delta f_{t_m}] \cdot T_{\text{frame}}, \\ \Delta t_c = [\Delta f_{t_1} + \Delta f_{t_2} + \cdots + \Delta f_{t_n}] \cdot T_{\text{frame}}, \end{cases} \tag{17}$$

where m and n are the numbers of pixels traversed along the row and column directions, respectively.

Given the angular resolution θ, array size of the detector $M \times M$, and the detection distance R, the physical coverage of a single pixel is

$$L_{\text{pixel}} = \tan\left(\frac{\theta}{M}\right) \cdot R. \tag{18}$$

The row and column velocity components are then expressed as

$$\begin{cases} v_r = \dfrac{m \cdot L_{\text{pixel}}}{\Delta t_r}, \\ v_c = \dfrac{n \cdot L_{\text{pixel}}}{\Delta t_c}, \end{cases} \tag{19}$$

and the motion direction is inferred from the temporal order of sub-frame photon arrivals. For example, if pixel A precedes pixel B ($f_s^A < f_s^B$), the trajectory proceeds from A to B. The terminal pixel (x_m, y_n) defines the endpoint for velocity vector $\mathbf{v}_{xy}$ estimation.

Within the terminal pixel, the photon arrival time at the final sub-frame $\hat{i} = F_{\text{res}}$ is refined using the temporal drift model from the two-stage RANSAC fitting,

$$\hat{t}_i = a \cdot F_{\text{res}} + b. \tag{20}$$

The slope a encodes the relative depth motion. By applying the time-of-flight relation with system delay t_0, the depth $\hat{R}_i$ is

$$\hat{R}_i = \frac{c}{2} \cdot \left(\hat{t}_i + t_0\right), \tag{21}$$

from which the depth velocity component is derived as

$$\vec{v}_z = \frac{\hat{R}_i - \hat{R}_{i-1}}{\Delta t_{\text{frame}}} = \frac{c}{2} \cdot \frac{a}{\Delta t_{\text{frame}}}. \tag{22}$$

2. Target Size Estimation from Pixel Overlap

When photon events appear simultaneously in adjacent pixels, the overlap duration encodes the projection of the physical size onto the detector plane (Fig. 5). The target width W and length L are estimated by

$$\begin{cases} W = v_c \cdot t_1, \\ L = v_r \cdot t_2, \end{cases} \tag{23}$$

where t_1 and t_2 are the coexistence durations in the column and row directions, respectively.

3. Sub-Pixel Rasterization for Super-Resolved Localization

To overcome the intrinsic resolution limit of the detector array, a sub-pixel rasterization strategy is introduced. After identifying the terminal pixel (x_m, y_n) and estimating its residence time

$$\Delta t^* = \left(f_e^* - f_s^* + 1\right) \cdot T_{\text{frame}}, \tag{24}$$

the intra-pixel displacement is calculated as

$$\begin{cases} \delta_x = v_c \cdot \Delta t^*, \\ \delta_y = v_r \cdot \Delta t^*. \end{cases} \tag{25}$$

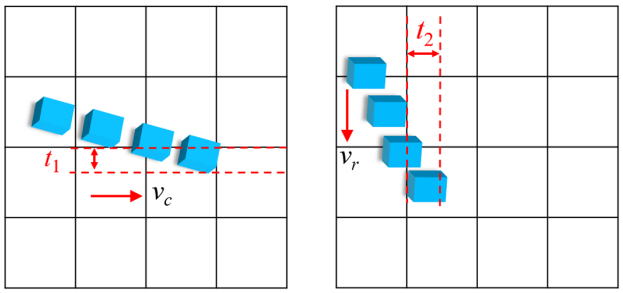


Fig. 5. Principle of target size estimation based on the overlap duration of photon events in neighboring pixels.

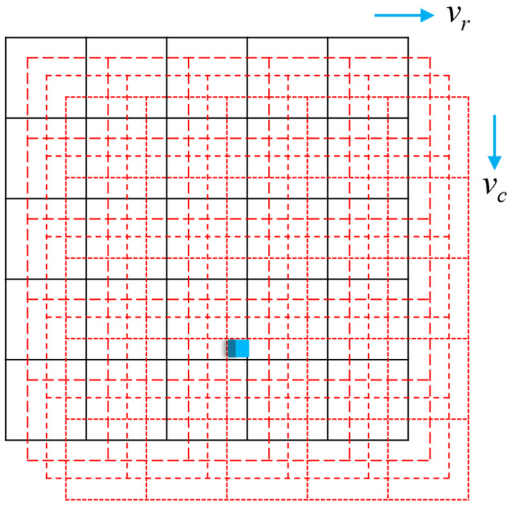


Fig. 6. Sub-pixel rasterization scheme for super-resolved localization.

Each pixel is partitioned into a 10 × 10 grid (Fig. 6), yielding a refined sub-pixel coordinate system. Considering the mismatch between the detector's active area and the geometric pitch (Fig. 7), target motion effectively scans dense trajectories across the pixel. This redundancy is exploited to suppress motion blur and enhance localization accuracy.

The final sub-pixel resolution is given by

$$\begin{cases} M_x = \text{Round}\left(\frac{\max(\delta_x, W)}{L_{\text{pixel}}}\right), \\ M_y = \text{Round}\left(\frac{\max(\delta_y, L)}{L_{\text{pixel}}}\right), \\ M_{\text{final}} = 10 \times \min(M_x, M_y), \end{cases} \tag{26}$$

where Round(·) denotes rounding to one decimal place. All indices and parameters used in the 5D-MR framework are listed in Table 2.

It is noted that the 5D-MR framework relies on detectable target motion across pixels within a single sampling interval to enable accurate velocity estimation. For instance, at a 10 Hz frame rate, a 1° × 1° FOV, a 64 × 64 detector array, and a 1 km imaging range, the minimum lateral motion required for reliable localization is approximately 1.366 m/s. Accordingly, the algorithm is inherently unsuitable for static or quasi-stationary targets.

4. MONTE CARLO SIMULATION VERIFICATION AND ANALYSIS OF GM-APD SPL DYNAMIC DETECTION

To rigorously assess the proposed 5D-MR framework, the dynamic far-field simulation environment was established using the Gm-APD triggering model combined with the Monte Carlo method [37], as illustrated in Fig. 8. The simulation workflow is as follows:

- Generating a 3D target trajectory from the kinematic model;
- Simulating signal and noise photons using the Gm-APD detection model;
- Integrating the trigger probability density function in discrete time to obtain the cumulative distribution function;
- Based on the inverse transform sampling method, the response time distribution of pixels is constructed and independently sampled to generate SPL data;
- Applying the proposed 5D-MR method for motion-blur suppression and super-resolution location.

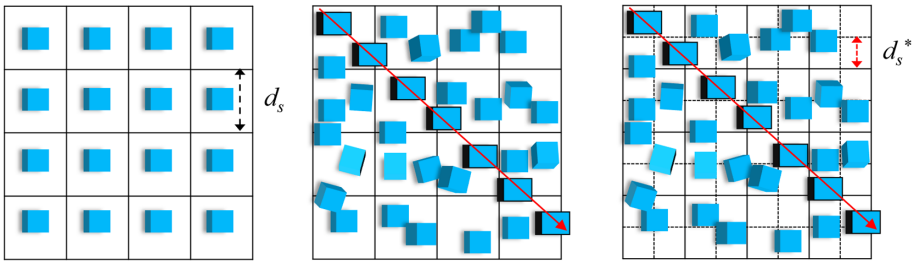


Fig. 7. Equivalent schematic of the detector array and effective motion scanning across the row / column.

Table 2. List of Indices and Parameters Used in the 5D-MR Framework

Index	Parameter / Variable	Index	Parameter / Variable	Index	Parameter / Variable
$T_{f,r,c}$	Photon arrival times at (r, c) in frame f	$N_{f,r,c}$	Photon events number at (r, c) in frame f	Ω_k	Set of pixels in k th sliding window
$H_k(t)$	Histogram of photon events in window k	$P_{r,c}$	Photon event ratio in peak interval	γ	Threshold for target pixel classification
$t_{\min}$, $t_{\max}$	Peak gating interval	$\mathcal{T}_{x,y}$	2D photon arrival-time sequence	a, b	Linear drift slope and intercept in RANSAC
ε_i	Measurement noise in ToF	σ	DBSCAN neighborhood radius	minPts	Minimum points threshold for DBSCAN
C_k	DBSCAN cluster	C^*	Main cluster of target photons	f_s, f_e	Start and end frame indices of main cluster
Δf	Frame span of target	$\hat{t}_i$	Refined photon arrival time	$\hat{R}_i$	Depth estimate from ToF
Δt_r, Δt_c	Total dwell times along row / column	m, n	Number of pixels along row / column	L_{pixel}	Physical coverage of a single pixel
v_r, v_c, v_z	Velocity along row, column, depth	t_1, t_2	Coexistence duration for width / length	W, L	Target width and length
δ_x, δ_y	Intra-pixel displacement	M_x, M_y	Sub-pixel grid size along x/y	M_{final}	Final sub-pixel resolution

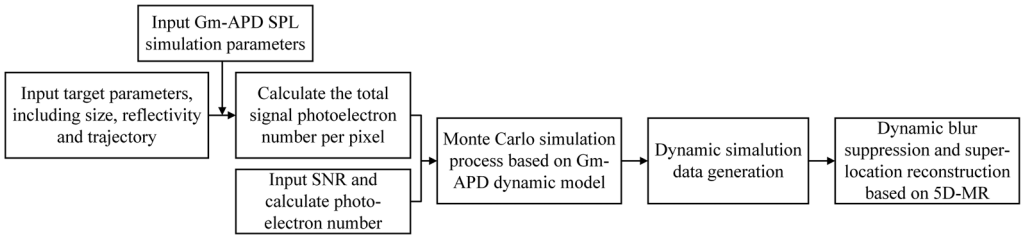


Fig. 8. Monte Carlo simulation framework for Gm-APD SPL dynamic detection.

The principal simulation parameters are listed in Table 3. Photon statistics account for optical aperture, quantum efficiency, system transmission, and atmospheric attenuation. A gating window of 1 μs was divided into 1000 temporal bins. The simulated target size corresponds to 0.5 pixel with a reflectivity of 0.3. Under these conditions, the mean detected signal photon number is $N_s = 0.5343$ per pixel, while noise levels are varied to achieve different SNR conditions.

The target moves with the constant velocity from initial position (x_0, y_0, z_0), where (x_0, y_0) lies on the FoV plane and z_0 is depth. The transverse and lateral velocities are $v_x = v_y = 15$ pixels/s, and the longitudinal velocity $v_z = 7$ m/s. Ignoring depth scaling, the simulated input trajectory is illustrated in Fig. 9. The size of the sliding window is carefully chosen to balance spatial photon aggregation and motion-induced smearing and must be jointly determined by the target dynamics and the reconstruction rate. In subsequent simulations, a window size of $W_s = 4$ is adopted. At a target range of 1 km, a single pixel corresponds to approximately 0.55 m laterally, so that a 4×4 window covers roughly 2.2 m. With a reconstruction rate of 5 Hz, this configuration limits the maximum allowable lateral target velocity to approximately 11 m/s at 1 km, ensuring that the target remains within the same sliding window between consecutive reconstructions and enabling stable photon accumulation.

Table 3. Simulation Parameters of Gm-APD SPL for Dynamic Detection

Parameter	Value
Laser wavelength	1064 nm
Laser repetition rate	2.5 kHz
FoV	1° × 1°
Optical efficiency	0.9
Atmospheric transmittance	1.3 km^{-1}
Detector pixels	64 × 64
Quantum efficiency	0.3
System efficiency	0.9
Gate width	1000 ns
Time resolution width	1 ns

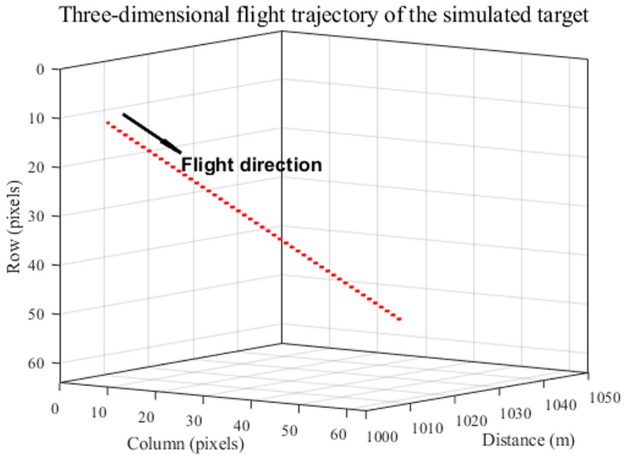
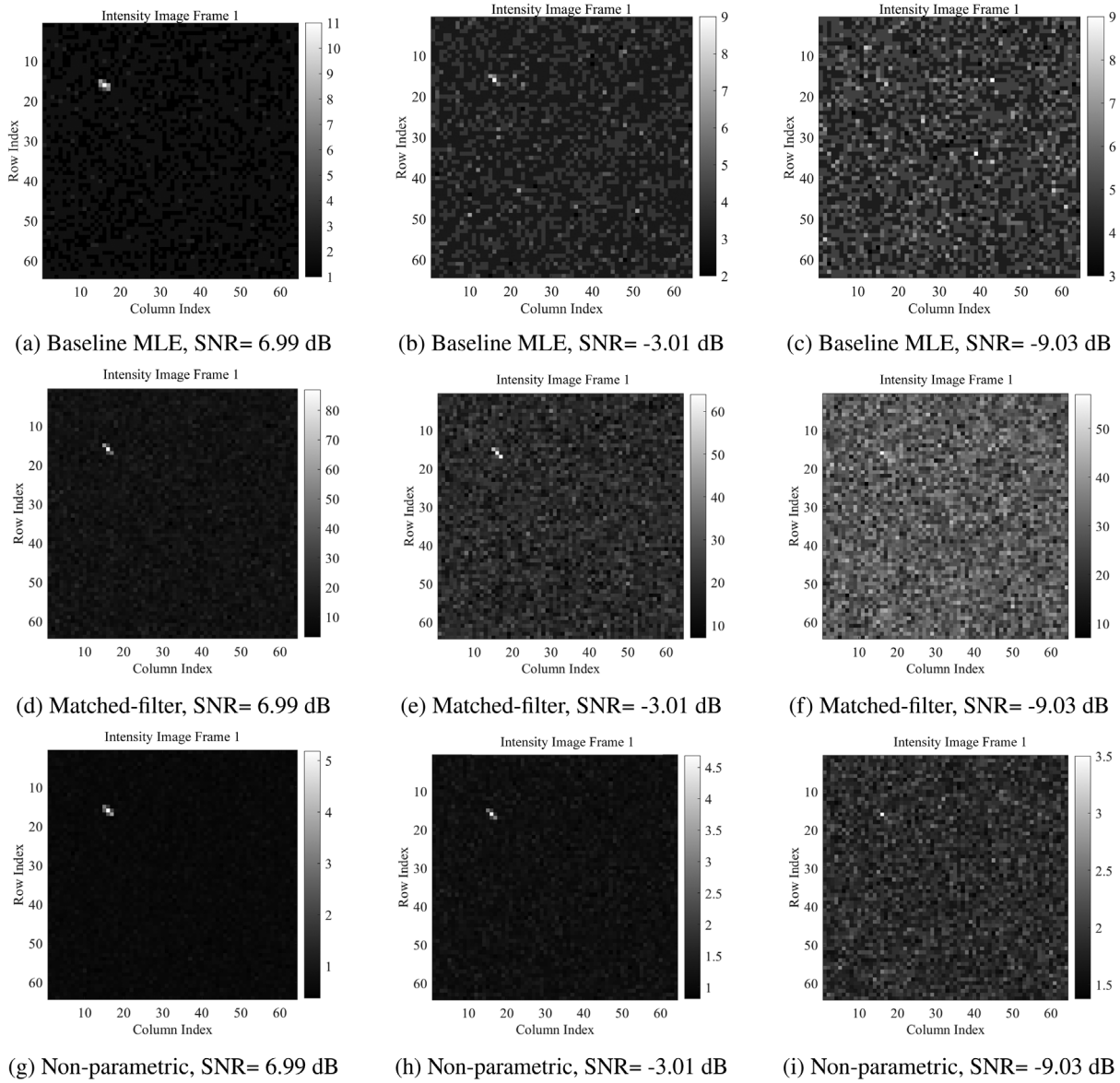


Fig. 9. 3D trajectory of the dynamic point target used as input in the simulation.

To evaluate the robustness of the proposed approach, datasets were generated with SNR values ranging from 6.99 down to −9.03 dB. Panel 10(a)–10(c) shows the reconstructed intensity using maximum likelihood estimation (MLE) [34] applied over 500 frames at a 5 Hz reconstruction rate. The corresponding matched-filter reconstruction results and nonparametric kernel density estimation (KDE) results are presented in panel 10(d)–10(f) and panel 10(g)–10(i), respectively. As the SNR drops below −3.01 dB, conventional approaches fail to resolve the target.

In contrast, signal preprocessing using the 5D-MR algorithm (Section 3.A) successfully identifies optimal local windows through kurtosis analysis and extracts ToF peaks even at SNR = −9.03 dB (Fig. 11). By comparison, conventional trajectory-based methods [21,26] fail when the SNR drops below 0 dB.

Continuous target motion during the reconstruction interval causes pronounced blur along the flight trajectory. Figure 12 illustrates the dual-domain RANSAC–DBSCAN photon-event

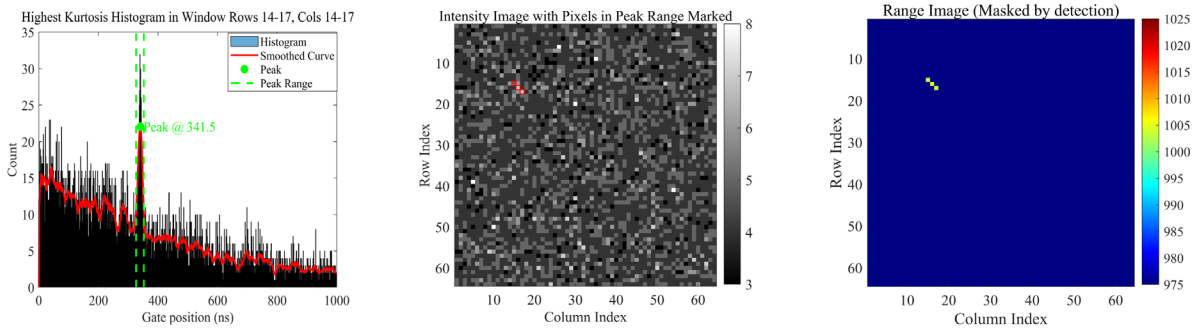


(a) Baseline MLE, SNR= 6.99 dB (b) Baseline MLE, SNR= -3.01 dB (c) Baseline MLE, SNR= -9.03 dB

(d) Matched-filter, SNR= 6.99 dB (e) Matched-filter, SNR= -3.01 dB (f) Matched-filter, SNR= -9.03 dB

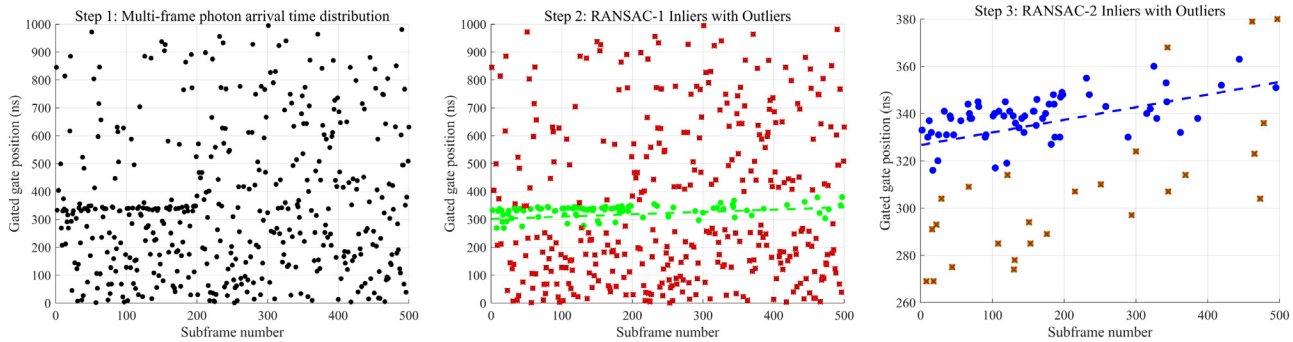
(g) Non-parametric, SNR= 6.99 dB (h) Non-parametric, SNR= -3.01 dB (i) Non-parametric, SNR= -9.03 dB

Fig. 10. Dynamic reconstruction results under different SNRs and reconstruction methods. Rows correspond to methods: (a)–(c) baseline MLE; (d)–(f) matched-filter reconstruction; (g)–(i) non-parametric estimation. Columns correspond to SNR conditions: left 6.99 dB, middle −3.01 dB, and right −9.03 dB.

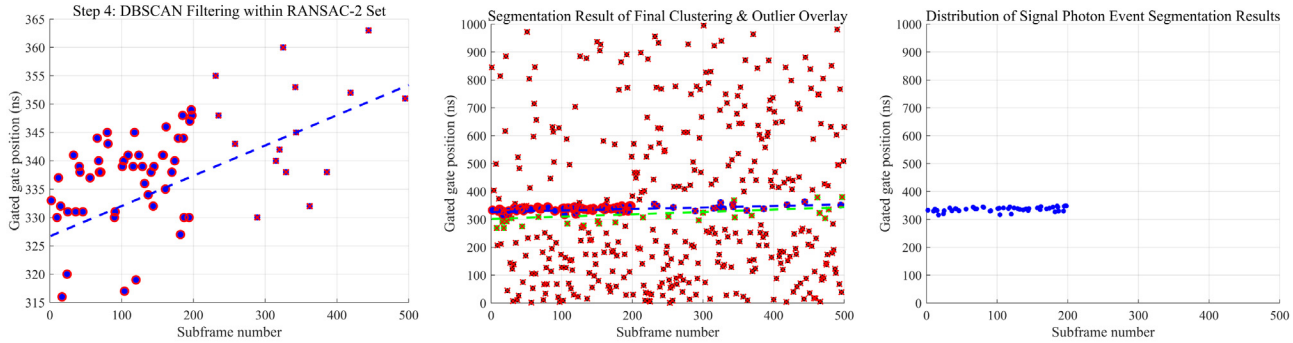


(a) Photon event frequency distribution (b) Detection result of intensity image (c) Detection result of range image

Fig. 11. Signal preprocessing and detection under the SNR of −9.03 dB.



(a) All photon events distribution (b) Inliers after first-stage RANSAC (c) Inliers after second-stage RANSAC



(d) DBSCAN-extracted main cluster (e) Segmented photon events (f) Target photon events distribution

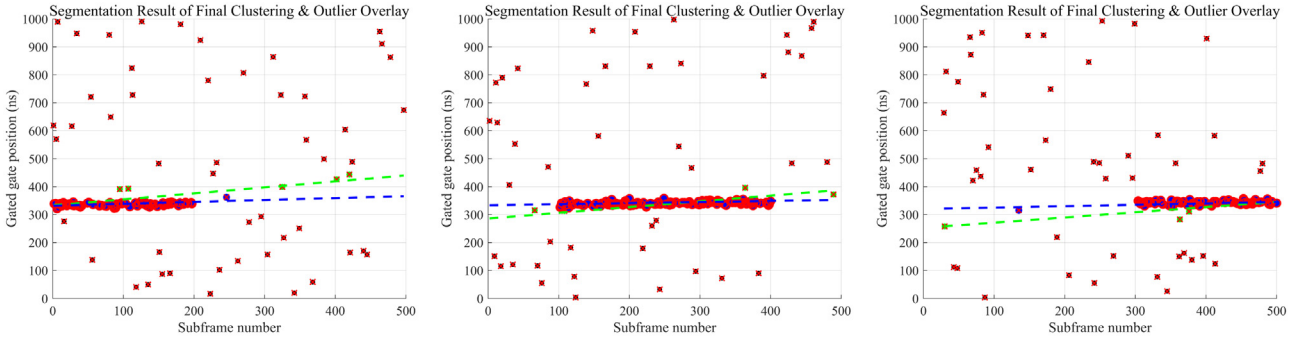
Fig. 12. RANSAC–DBSCAN photon-event segmentation process at SNR = −9.03 dB. (Black markers denote all photon events detected within the target region over the sampling interval. Green markers represent inliers identified by the first-stage RANSAC, with green dashed lines indicating the corresponding fitted trajectories. Blue markers denote inliers retained after the second-stage RANSAC, and blue dashed lines show the final fitted photon-event distribution. Black markers marked with red crosses indicate outlier photon events rejected after the second-stage RANSAC, whereas black markers circled in red correspond to target photon events belonging to the dominant cluster extracted by DBSCAN within the RANSAC inlier set. The same marker and line-style conventions are used in all subsequent panels.)

segmentation (Section 3.B) for pixel (15,15) at SNR = −9.03 dB, recovering the target's true position. The horizontal axis denotes the sub-frame index, and the vertical axis the gated photon arrival time. Panel 12(a) shows all photon events (black dots); panels 12(b) and 12(c) show inliers from first- and second-stage RANSAC (green and blue circles) with linear fits (dashed lines). Panel 12(d) displays the DBSCAN-extracted main cluster (red circles) representing target photons, while outliers are marked by red crosses. Finally, the distribution of photon events triggered by the target remains as shown in panel 12(f).

Photon-event segmentation results under different SNR conditions are shown in Figs. 13–15. At the minimum detectable SNR of −9.03 dB, Table 4 summarizes the signal photon statistics for three representative pixels.

From Table 4, the target path is inferred from (15,15) to (17,17), with terminal localization in pixel (17,17). Second-stage RANSAC regression yields $y = 0.0175x + 331.4386$, corresponding to outward motion. Substituting the final frame index $i = 500$ into the fitted model results in a corrected ToF of 340.211 ns, corresponding to a target range of 1006.031 m, as shown in Fig. 16, implying a longitudinal velocity $v_z = 6.563$ m/s. Compared with the uncorrected range of 1003.572 m, the error decreases from 2.03 to 0.43 m.

Meanwhile, the target fully traverses pixel (16,16) with a dwell of 298 frames, overlapping 96 frames with the subsequent pixel.

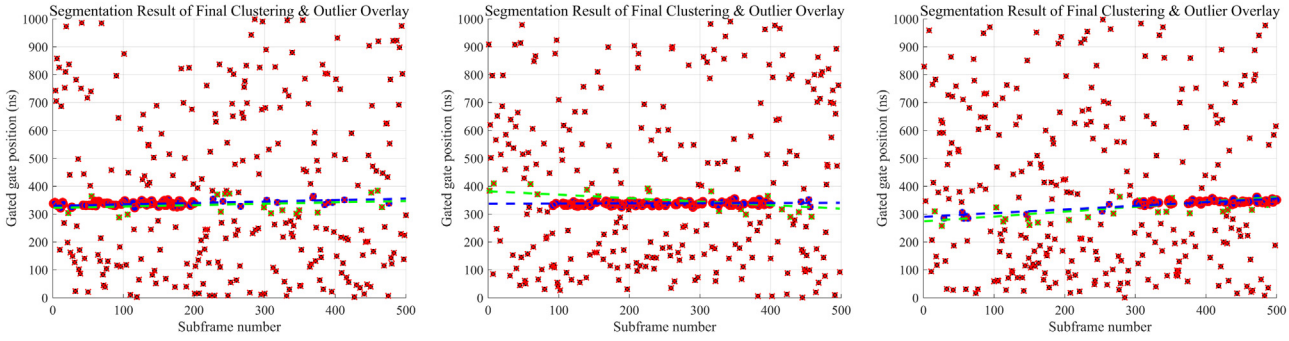


(a) Segmentation result of pixels (15,15)

(b) Segmentation result of pixels (16,16)

(c) Segmentation result of pixels (17,17)

Fig. 13. 5D-MR result of photon event segmentation at SNR = 6.99 dB.

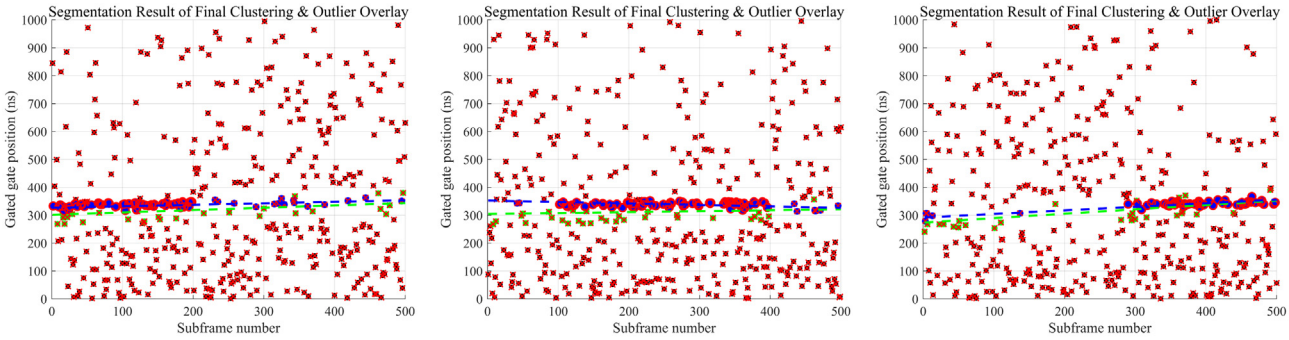


(a) Segmentation result of pixels (15,15)

(b) Segmentation result of pixels (16,16)

(c) Segmentation result of pixels (17,17)

Fig. 14. 5D-MR result of photon event segmentation at SNR = −3.01 dB.



(a) Segmentation result of pixels (15,15)

(b) Segmentation result of pixels (16,16)

(c) Segmentation result of pixels (17,17)

Fig. 15. 5D-MR result of photon event segmentation at SNR = −9.03 dB.

Table 4. Frame Parameters of Triggering Photon Events of Detected Pixels under SNR = −9.03 dB

Pixel Position	Start Frame	End Frame	Frame Span
(15,15)	2	199	198
(16,16)	101	398	298
(17,17)	303	498	196

The corrected photon arrival time for pixel (16,16) is 339.479 ns, corresponding to a target range of 1005.922 m. Considering the system's angular resolution, the spatial footprint of a single pixel is estimated as 0.273 m × 0.273 m. Using Eq. (19), the target's row and column velocities are then determined as

$$v_r = v_c = \frac{L_{\text{pixel}}}{\Delta t_r} = 3.362\ \text{s}. \tag{27}$$

The co-occurrence duration between two adjacent pixels indicates that the target simultaneously occupies the spatial regions of both pixels during this interval. Based on Eq. (23), the physical dimensions of the target can be estimated as $W = L = 0.129$ m.

For the final pixel (17,17), the target dwells for 196 frames. Accordingly, Eq. (25) is used to compute the target's displacement along the row and column directions within this pixel:

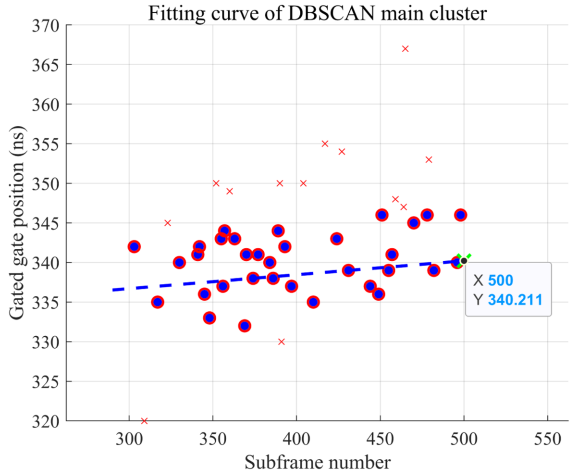


Fig. 16. Fitting result of the photon-event distribution at the corrected pixel (17,17).

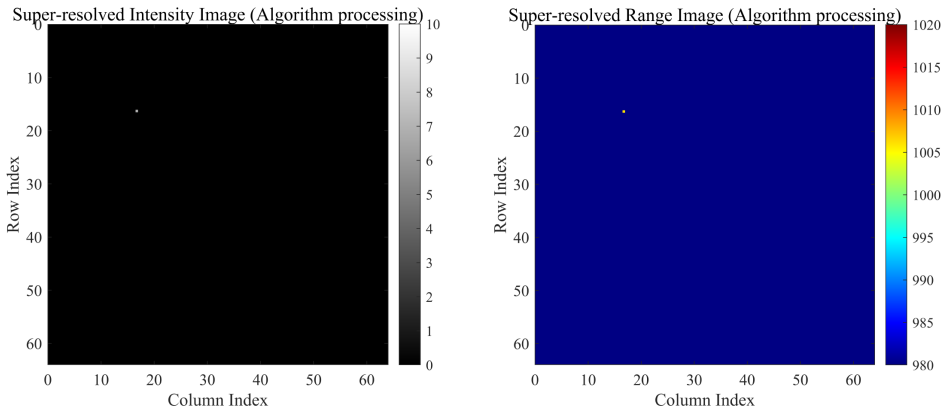
$$\delta_x = \delta_y = v_c \cdot \Delta t^* = 0.263\ \text{m}. \tag{28}$$

Considering the corrected range of 1006.031 m for this pixel, the corresponding spatial footprint is 0.273 m × 0.273 m. Substituting into Eq. (26) provides the super-resolution imaging positions, as illustrated in Figs. 17 and 18, with a reconstruction accuracy better than 0.1 pixel.

Table 5 summarizes the simulated input (ground-truth) target ranges, the reconstruction results from conventional MLE, and the corrected target ranges obtained using multi-frame 5D-MR processing at SNR = −9.03 dB. Quantitative analysis of the reconstruction errors highlights the significant advantage of 5D-MR [38]: MLE exhibits a mean range error of −6.672 m with a standard deviation of 3.178 m, whereas 5D-MR achieves a mean error of 0.202 m and a standard deviation of 0.211 m. This corresponds to an approximately one-order-of-magnitude reduction in both bias and dispersion of the range estimation error, demonstrating markedly improved accuracy and stability under extremely photon-starved conditions.

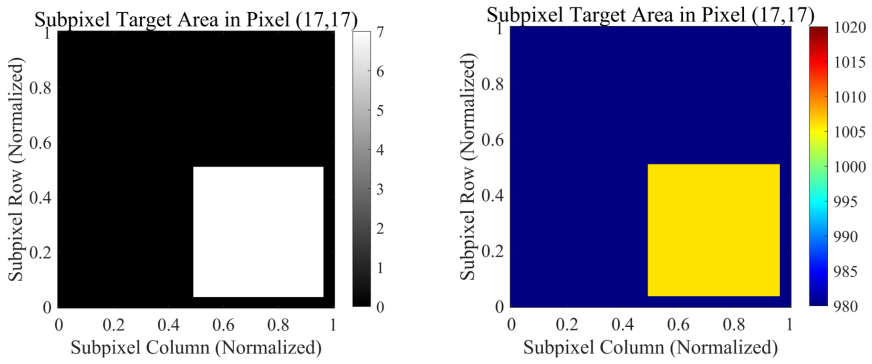
In the above simulations, the target is assumed to undergo linear motion. However, when the target follows a curved trajectory in three-dimensional space (e.g., during turning maneuvers), its projection onto the detector plane exhibits pronounced nonlinear motion characteristics. Under the same simulation parameters, the corresponding signal preprocessing and detection results for the nonlinearly moving target are shown in Fig. 19, where the target is ultimately detected across three neighboring pixels, namely (35,13), (35,12), and (34,12).

The corresponding dual-domain RANSAC–DBSCAN photon-event segmentation and fitting results are presented in Fig. 20. Based on the spatial distribution of photon events accumulated toward the end of the sampling period, the target is uniquely localized to the detector pixel (34,12). Following the same super-resolution estimation procedure described above, the sub-pixel target position at the end of the sampling interval can be readily obtained. For brevity, the detailed calculation is not repeated here.



(a) Reconstruction of intensity image (b) Reconstruction of range image

Fig. 17. Final reconstruction result after the 5D-MR processing.

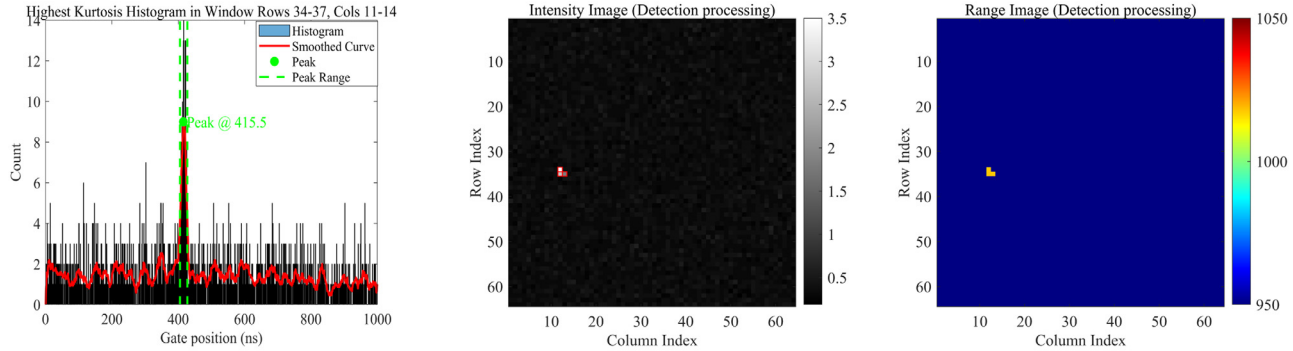


(a) Reconstruction of intensity image (b) Reconstruction of range image

Fig. 18. Reconstruction result after pixel enlargement.

Table 5. Range Errors for Different Reconstruction Methods over Consecutive Reconstructed Frames at SNR = −9.03 dB

Index	1	2	3	4	5	6	7	8
Simulated true value (m)	1005.601	1007.305	1008.598	1010.204	1011.821	1013.302	1014.871	1016.343
MLE reconstruction (m)	1003.572	1002.252	1005.526	1003.508	1004.027	1005.386	1005.653	1004.745
5D-MR corrected (m)	1006.031	1007.429	1008.839	1010.682	1012.178	1013.298	1014.758	1016.449
Range error of MLE (m)	−2.029	−5.053	−3.072	−6.696	−7.794	−7.916	−9.218	−11.598
Range error of 5D-MR (m)	0.430	0.124	0.241	0.478	0.357	−0.004	−0.113	0.106

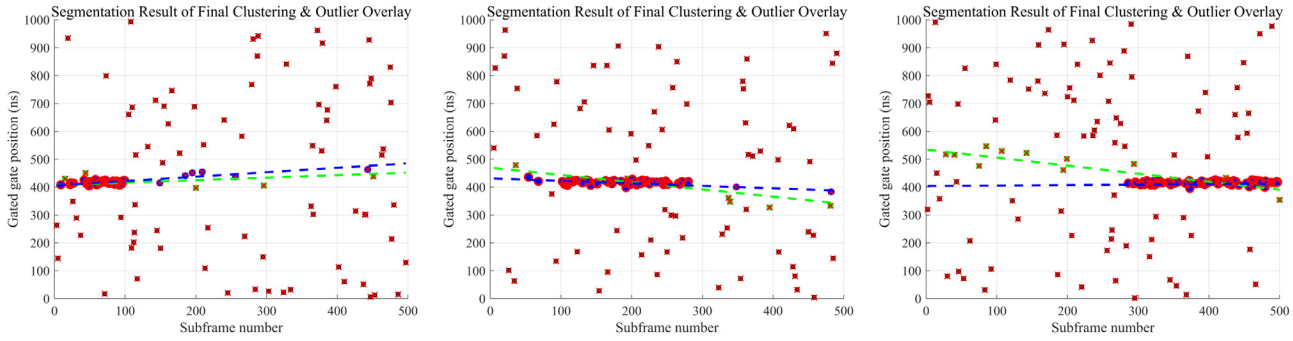


(a) Photon event frequency distribution under nonlinear target motion

(b) Intensity-image detection result for nonlinear motion

(c) Range-image detection result for nonlinear motion

Fig. 19. Signal preprocessing and detection results for a nonlinearly moving target.



(a) Photon event segmentation at pixel (35,13) under nonlinear motion

(b) Photon event segmentation at pixel (35,12) under nonlinear motion

(c) Photon event segmentation at pixel (34,12) under nonlinear motion

Fig. 20. Photon-event segmentation results for a nonlinearly moving target.

While the simulations demonstrate the capability of 5D-MR to accurately reconstruct both linear and nonlinear target motion under photon-starved conditions, practical deployment on embedded SPL systems requires consideration of computational feasibility. To this end, we performed a quantitative complexity analysis comparing the proposed 5D-MR framework against typical deep neural network (DNN) architectures used for LiDAR point cloud processing (e.g., 3D-CNNs [39] or PointNet++ [40]). The computational cost is estimated based on floating point operations (FLOPs) per detection frame. We define the system parameters consistent with our simulated experimental setup:

- The focal-plane array of the detector has a spatial resolution of 64×64 pixels;
- Temporal gate depth is 1000 ns;
- Sliding window size: $W \times W = 4 \times 4$;
- Photon density: under low-SNR conditions with multi-frame accumulation (500 frames), the average number of photon events triggered per pixel is approximately 450, yielding a total event count $N_{\text{total}} \approx 1.84 \times 10^6$.

The computational load of 5D-MR is dominated by two stages: the sliding-window histogram construction and the kurtosis-based statistical scanning. The complexity can be approximated as

$$C_{\text{5D-MR}} \approx \underbrace{N_{\text{total}} \cdot W^2}_{\text{Histogram Update}} + \underbrace{N_{\text{pix}} \cdot N_{\text{bins}} \cdot \eta}_{\text{Statistical Metrics}}, \tag{29}$$

where η represents the atomic operations for calculating moments (mean, variance, and kurtosis) per bin ($\eta \approx 15$). Substituting the parameters, the histogram construction requires $\approx 2.94 \times 10^7$ operations, and the statistical scanning requires $\approx 6.14 \times 10^7$ operations. The subsequent RANSAC and DBSCAN steps operate on a sparse subset of candidate points, contributing negligibly to the total load:

$$C_{\text{5D-MR}}^{\text{total}} \approx 9.08 \times 10^7 \text{ FLOPs} \approx 0.09 \text{ GFLOPs}. \tag{30}$$

For comparison, we consider a lightweight 3D-CNN processing the spatio-temporal volume ($H \times W \times T$). Even with a simplified architecture (e.g., a single feature extraction layer with $3 \times 3 \times 3$ kernels and 16 filters), the convolution operations across the voxel grid are computationally expensive. A conservative lower bound for the inference cost of such volumetric networks is typically cited in the range of gigas-of-operations [41,42]:

$$C_{\mathrm{NN}}^{\mathrm{total}} \geq 2.5 \times 10^9 \ \mathrm{FLOPs} = 2.5 \ \mathrm{GFLOPs}. \tag{31}$$

Table 6 summarizes the comparison results. The proposed 5D-MR achieves a speedup ratio of approximately 27× compared to the learning-based baseline. Furthermore, 5D-MR is a parameter-free approach guided by physical models, whereas neural networks typically require storing millions of weight parameters, imposing significant memory pressure on edge devices.

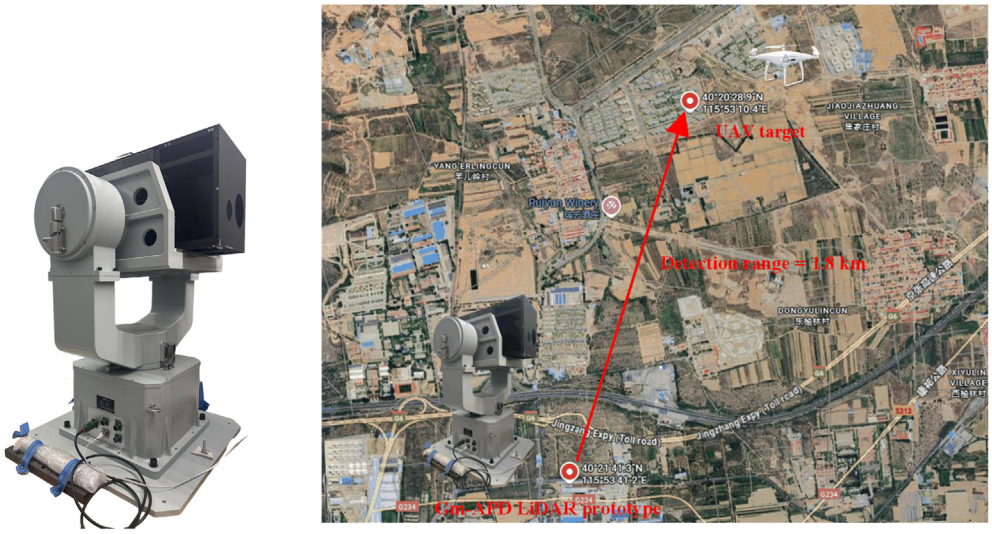
Table 6. Quantitative Comparison of Computational Complexity per Imaging Frame

Metric	Proposed 5D-MR	Typical 3D-CNN / PointNet
FLOPs	∼0.09 GFLOPs	∼2.5 GFLOPs
Learnable parameters	0 (model-based)	$> 10^6$ (data-driven)
Memory footprint	Linear w.r.t raw data	High (weights + feature maps)

5. FIELD EXPERIMENT VALIDATION AND RESULTS ANALYSIS

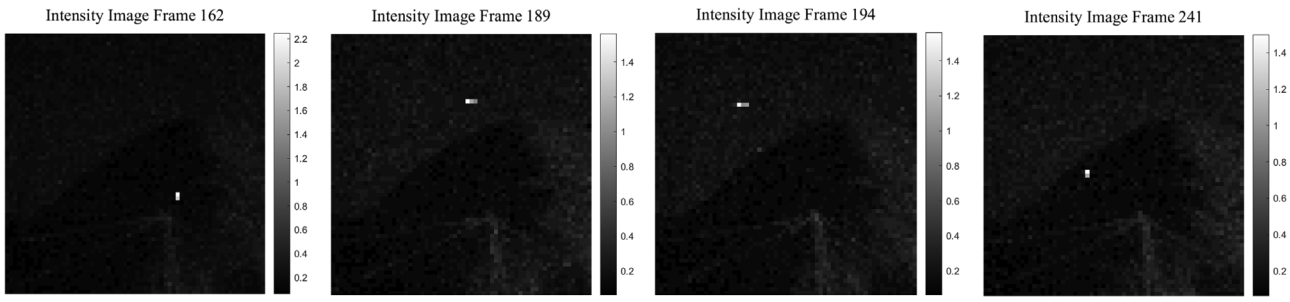
To assess the performance of the proposed algorithm in realistic scenarios, we conducted a far-field experiment using a Gm-APD-based SPL system. The campaign was carried out in Yanqing District, Beijing, China (40.41°N, 115.53°E), under visibility of approximately 31 km and ambient illuminance exceeding 113,500 lux, representing a highly challenging photon-starved environment. Figure 21 illustrates the experimental system and scene.

The system employed a pulsed laser with a wavelength of 1064 nm, a repetition rate of 2.5 kHz, a pulse width of 10 ns, and an average output power of 6.8 W, corresponding to a single-pulse energy of 2.72 mJ. The transmitted laser illuminated an FoV of 1° × 1°, and the backscattered photons were collected by a receiver with an 800 mm aperture, equipped with a 64 × 64 Gm-APD focal plane array. The system operated with a gating width of 2000 ns and a time resolution of 1 ns. The target was a DJI Phantom 4 UAV flying at a slant range of 1.8 km and an altitude of approximately 45 m, with transverse and longitudinal velocities of about 12 and 5 m/s, respectively. The UAV dimensions were 289.5 mm × 289.5 mm × 180 mm. Each imaging sequence was



(a) Gm-APD SPL prototype (b) Experimental scene diagram

Fig. 21. Schematic diagram of detecting a far-field UAV target.



(a) Frame 162 (b) Frame 189 (c) Frame 194 (d) Frame 241

Fig. 22. Reconstruction result of the far-field UAV target (intensity profile).

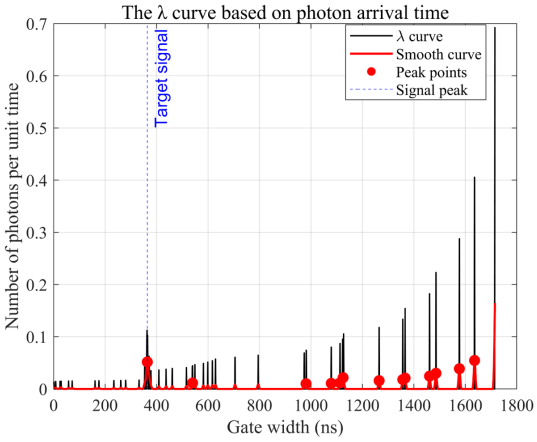


Fig. 23. Average photon distribution in the target pixels.

reconstructed by integrating 500 consecutive frames, yielding an effective frame rate of 5 Hz, as shown in Fig. 22.

Adaptive noise filtering was applied to suppress spurious triggers at the gate edges. As shown in Fig. 23, the average number of detected signal photons per pixel was only 0.3550, whereas the mean background photons per pixel reached 2.7977, corresponding to an SNR of −8.97 dB. Under such conditions, conventional reconstruction exhibited severe motion blur and failed to recover reliable target images, highlighting the necessity of the proposed method.

In the transverse scenario, the reconstructed intensity images [Fig. 22(c)] exhibited several adjacent pixels with comparable photon amplitudes. However, the spatial photon distribution was highly irregular, and the intensity magnitude alone was insufficient to reliably determine the UAV's true location. After applying the proposed signal preprocessing and target detection algorithm as shown in Fig. 24, the reconstructed intensity image [Fig. 24(b)] appeared to indicate that the UAV occupied three adjacent pixels with clearly enhanced intensity. However, photon-count statistics revealed that four pixels were in fact associated with the target's motion. This discrepancy highlights that intensity magnitude alone cannot reliably represent the true spatial extent of the dynamic target, necessitating further analysis based on photon-event segmentation.

After detecting the candidate target pixels, photon-event segmentation was performed to resolve the ambiguity inherent in the intensity image, as shown in Fig. 25. By analyzing the temporal distribution of photon occurrences across consecutive frames and quantifying their co-occurrence through statistical Table 7, the UAV's motion trajectory was determined. The results indicate that the UAV migrated from pixel (18,23) to (18,20), dwelling for 168 frames at (18,21) with a 60-frame overlap with adjacent pixels. The overall trajectory spanned 108 frames. Based on the system's spatial resolution of 0.490 m × 0.490 m at a range of 1.8 km, the transverse velocity was estimated as 11.34 m/s. The 60-frame co-occurrence corresponds to a physical transverse width of 0.272 m, while the 87-frame dwell in the terminal pixel (18,20) indicates a flight distance of 0.395 m.

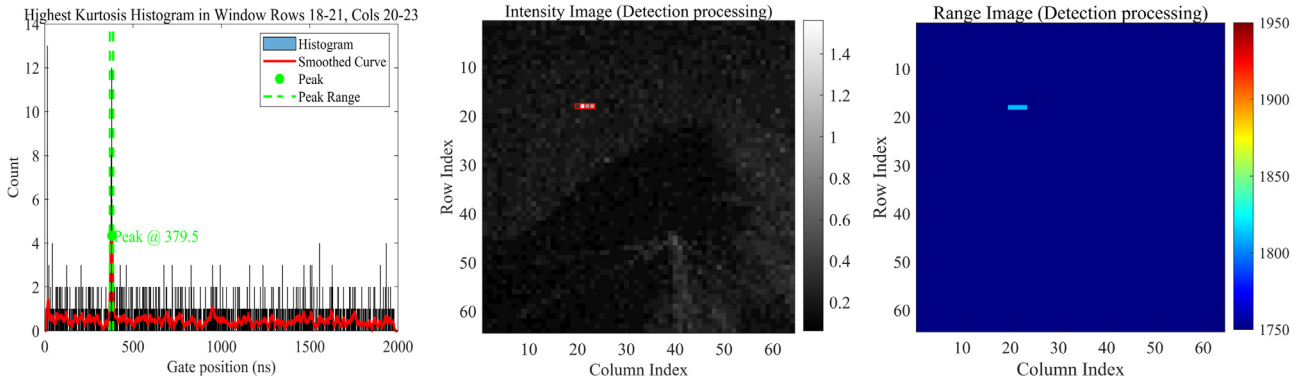
Table 7. Statistics of Signal Photon Events Segmentation for Detected Pixels

Pixel Position	Start Frame	End Frame	Frame Span
(18,23)	4	148	145
(18,22)	35	233	199
(18,21)	283	470	168
(18,20)	399	496	87

In the longitudinal scenario [Fig. 22(a)], after applying the proposed signal preprocessing and target detection algorithm as shown in Figs. 26 and 27, the reconstructed intensity image suggested that the UAV occupied only a few neighboring pixels with enhanced amplitude. However, photon-count statistics revealed that additional pixels were associated with the target's motion, indicating that intensity magnitude alone was insufficient to determine the true location.

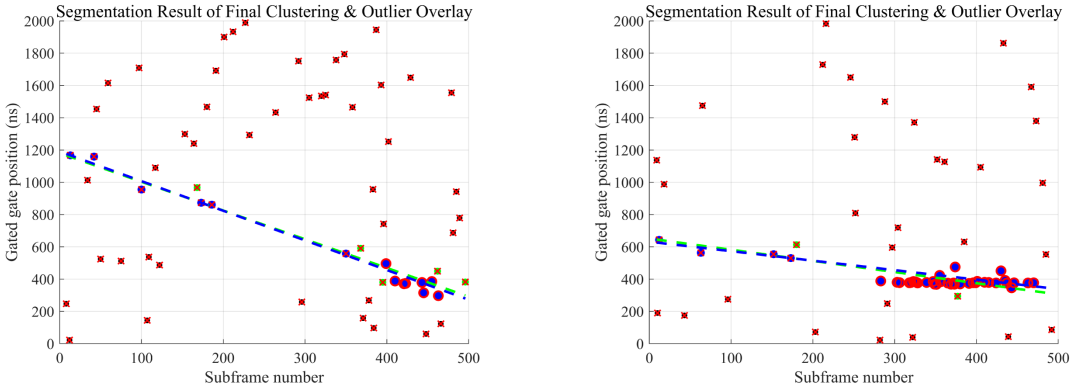
After detecting the candidate pixels of the two frames, photon-event segmentation was performed to accurately resolve the target trajectory, as shown in Fig. 28.

By analyzing the temporal distribution of photon events across consecutive frames and quantifying their co-occurrence using statistical analysis in Table 8, the UAV was found to migrate from pixel (41,43) to (39,43), ultimately stopping at (39,43). During this process, the UAV dwelled for 470 frames at pixel (40,43), overlapping with adjacent pixels for 205 frames, yielding a total coverage of 265 frames. Using the system's spatial resolution of 0.490 m × 0.490 m at a range of 1.8 km, the longitudinal velocity was calculated as 4.62 m/s. The co-occurrence segment corresponded to a longitudinal physical size of 0.214 m, while the 313-frame dwell in the terminal pixel (39,43) indicated a flight distance of 0.327 m.



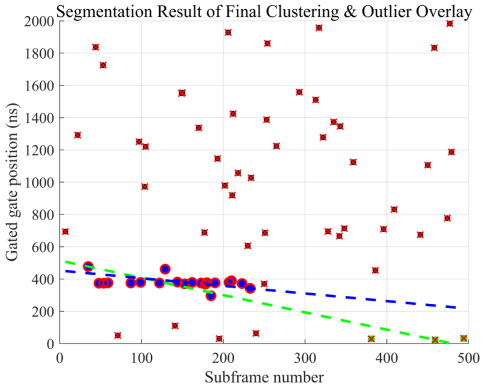
(a) Photon event frequency distribution (b) Detection result of intensity image (c) Detection result of range image

Fig. 24. Signal preprocessing and detection processing results.

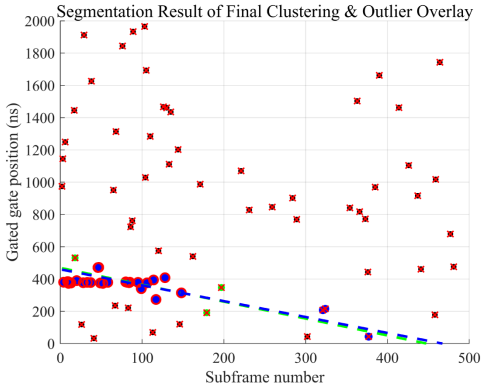


(a) Photon events segmentation of pixel (18,20)

(b) Photon events segmentation of pixel (18,21)

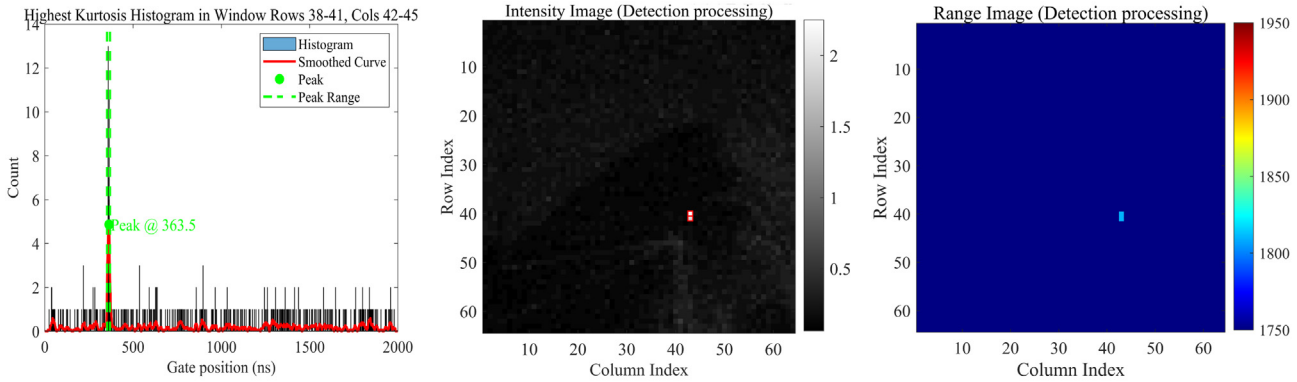


(c) Photon events segmentation of pixel (18,22)



(d) Photon events segmentation of pixel (18,23)

Fig. 25. Photon-event segmentation and motion-blur suppression results in the transverse flight experiment.



(a) Photon event frequency distribution (b) Detection result of intensity image (c) Detection result of range image

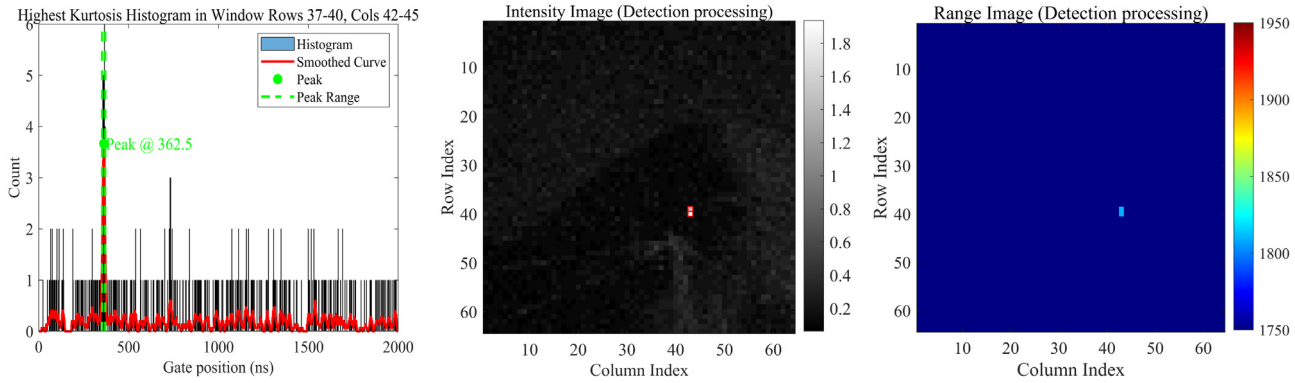
Fig. 26. Signal preprocessing and detection processing results of Frame 162.

Table 8. Parameter Statistics of Segmented Photon Events

Frames	Pixel Position	Start Frame	End Frame	Frame Span
162	(41,43)	3	337	335
	(40,43)	218	498	281
163	(40,43)	1	381	381
	(39,43)	187	500	314

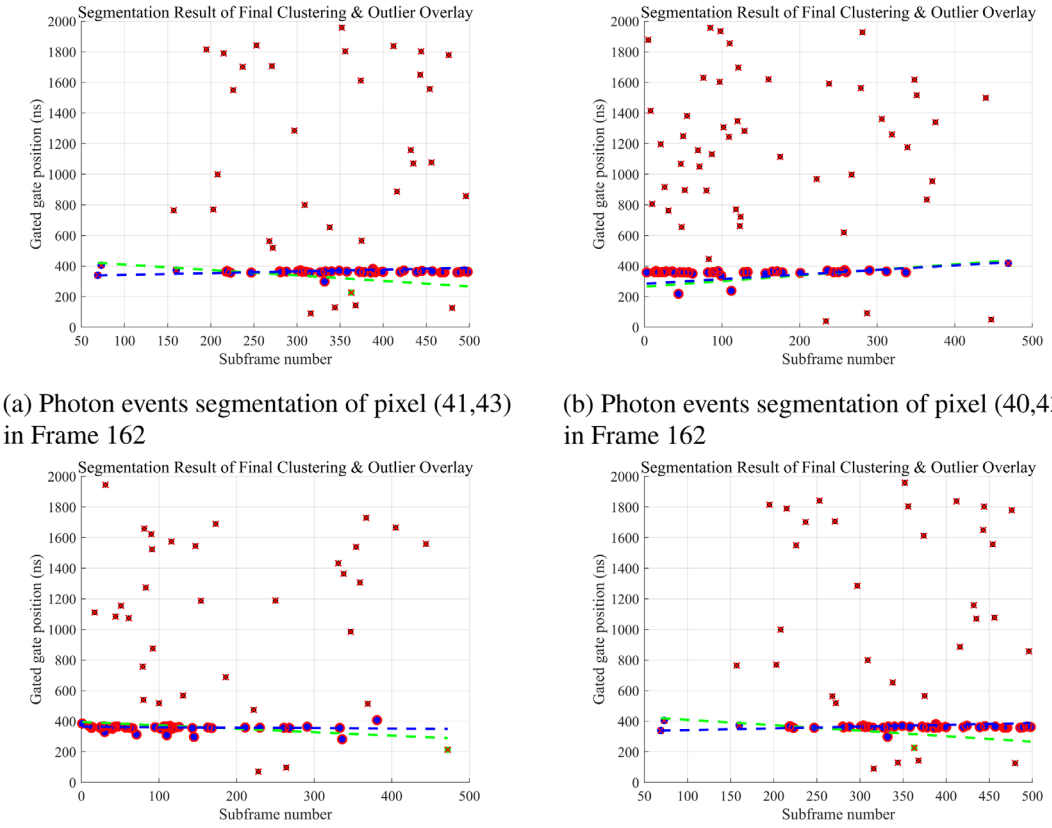
The above analyses show that intensity images alone cannot reliably reveal either the true trajectory or the physical dimensions of the dynamic UAV target. By integrating photon-event segmentation with temporal co-occurrence statistics, the target's two-dimensional physical size was estimated as approximately 0.272 m × 0.214 m. Building on this information, motion-blur suppression and super-resolved reconstruction were performed using the terminal-pixel dwell statistics, yielding high-quality intensity and depth images for both the transverse and longitudinal scenarios (Fig. 29).

To quantitatively validate the localization accuracy, the results were compared against a co-axially aligned visible camera with a FoV of 4.06° × 2.29°, a resolution of 3840 × 2160, and an angular resolution of 0.001058° per pixel. Unlike conventional reconstruction methods, whose localization accuracy is inherently



(a) Photon event frequency distribution (b) Detection result of intensity image (c) Detection result of range image

Fig. 27. Signal preprocessing and detection processing results of Frame 163.



(a) Photon events segmentation of pixel (41,43) in Frame 162

(b) Photon events segmentation of pixel (40,43) in Frame 162

(c) Photon events segmentation of pixel (40,43) in Frame 163

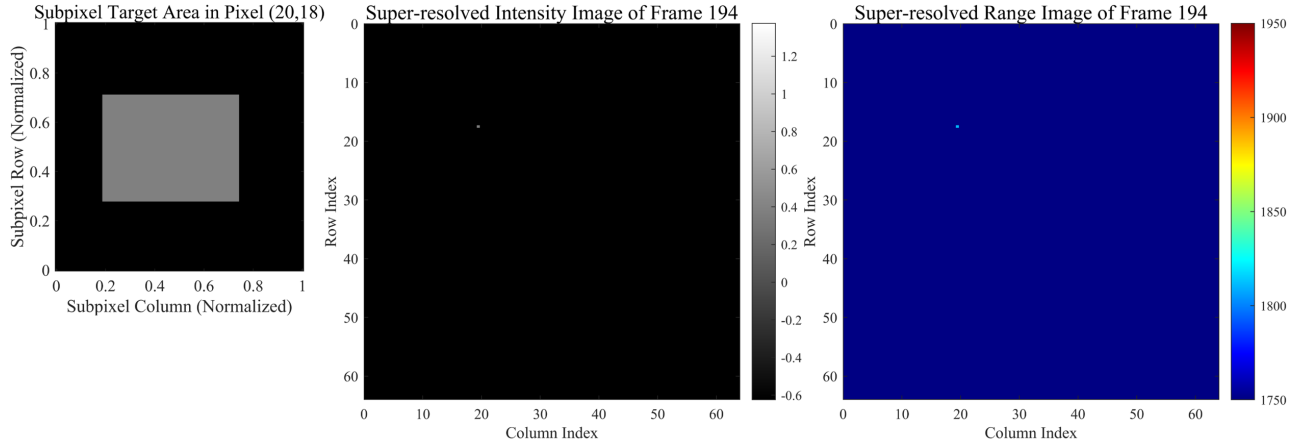
(d) Photon events segmentation of pixel (39,43) in Frame 163

Fig. 28. Photon-event segmentation and motion-blur suppression results in the longitudinal flight experiment.

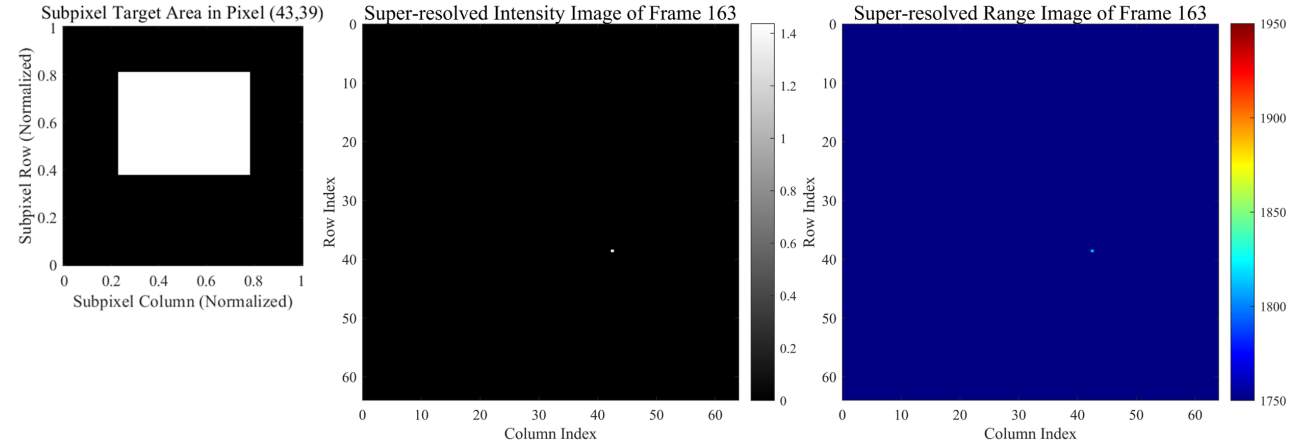
limited by the detector pixel scale, the proposed 5D-MR framework overcomes this constraint by exploiting the spatiotemporal correlation of photon events, achieving super-resolved localization beyond the native pixel array resolution. As shown in Figs. 30 and 31, the reconstructed UAV positions obtained from the SPL using the proposed 5D-MR algorithm exhibit high consistency with the optical reference captured by the visible camera. Specifically, in each figure, panel (a) presents the super-resolved localization results processed by the 5D-MR algorithm, panel (b) shows the corresponding co-axial camera image, and panel (c) displays the magnified view of the UAV target region.

6. CONCLUSION

We have presented the five-dimensional motion-aware reconstruction (5D-MR) framework, which suppresses motion blur and achieves super-resolved localization of dynamic UAV targets under ultra-low SNR conditions using SPL. By jointly exploiting time-of-flight, two-dimensional spatial coordinates, per-pixel photon statistics, and inter-frame continuity within local sliding windows, the method departs from conventional image-domain processing and enables robust photon extraction, trajectory estimation, and sub-pixel localization.

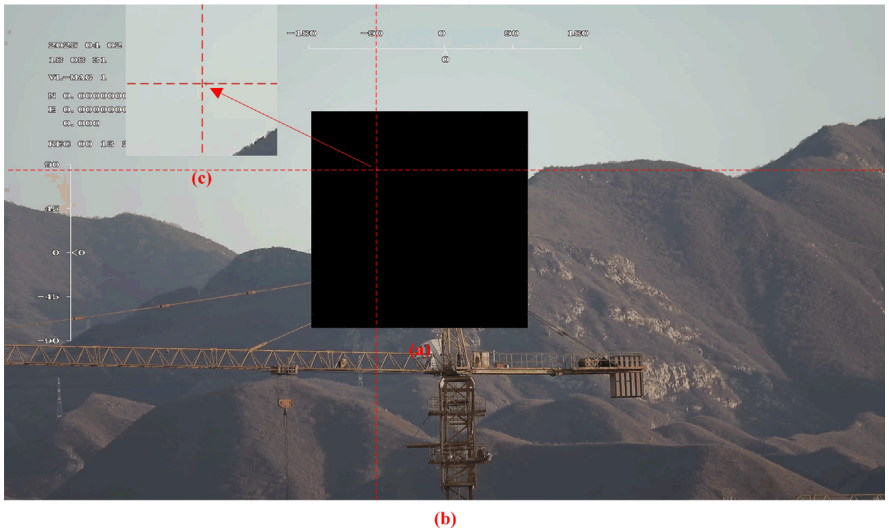


(a) Reconstruction results of transverse flight of Frame 194



(b) Reconstruction results of longitudinal flight of Frame 163

Fig. 29. Final super-resolved reconstruction result after algorithm processing.



(b)

Fig. 30. Coaxial imaging results of camera and SPL processed by the 5D-MR algorithm (transverse flight). (a) The super-resolved localization results processed by the 5D-MR algorithm; (b) the corresponding co-axial camera image; (c) the magnified view of the UAV target region.

Both Monte Carlo simulations and controlled field trials validate the effectiveness of the framework. At SNR levels around −9 dB, with mean signal returns below 0.6 photons per pixel, 5D-MR achieves angular errors within 0.1° and range errors under 0.5 m in simulation, and angular precision better than 0.01 pixel with size-estimation errors below 0.02 m in a 1.8 km field

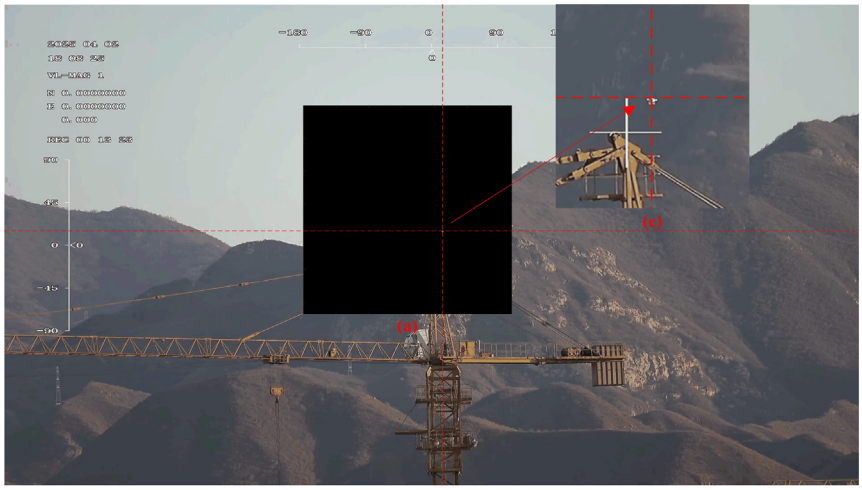


Fig. 31. Coaxial imaging results of the camera and SPL processed by the 5D-MR algorithm (longitudinal flight). (a) The super-resolved localization results processed by the 5D-MR algorithm; (b) the corresponding co-axial camera image; (c) the magnified view of the UAV target region.

experiment. Compared with the intrinsic angular resolution of the SPL system, these results represent nearly two orders of magnitude improvement in localization precision. These results confirm the feasibility of extending SPL imaging into the photon-starved regime where conventional methods fail.

The applicability of 5D-MR is, however, bounded by several conditions. First, it requires cross-pixel motion during the sampling interval; with typical array size, FoV, and frame-rate settings, this corresponds to a minimum effective target velocity of roughly 1–2 m/s. The method is therefore not suitable for static or hovering targets. Second, the temporal fitting assumes near-linear motion within the sliding window, and its accuracy deteriorates under highly non-linear or abrupt maneuvers. Third, in multi-target or swarm scenarios with intersecting trajectories, clustering and fitting may suffer from trajectory mixing, requiring more advanced data association. In addition, performance is ultimately limited by system calibration, clock precision, and extreme background illumination, and the computational load may challenge real-time deployment without hardware acceleration.

Overall, 5D-MR is well suited for long-range, sparse, dynamic target detection under ultra-low SNR, where cross-pixel motion provides redundancy for motion compensation and super-resolution. Future work will focus on adaptive gating and parameter tuning, fusion with complementary sensing modalities such as optics or radar, and extension to multi-target and wide-area monitoring to further broaden the operational scope of photon-efficient SPL imaging.

Funding. National Natural Science Foundation of China (62505069).

Acknowledgment. The authors would like to acknowledge the support of the National Key Laboratory of Laser Spatial Information, Harbin Institute of Technology.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

REFERENCES

1. J. Wang, Y. Liu, and H. Song, "Counter-unmanned aircraft system(s) (C-UAS): state of the art, challenges, and future trends," IEEE Aerosp. Electron. Syst. Mag. **36**(3), 4–29 (2021).
2. H. Wang and J. Wang, "Enhancing multi-UAV air combat decision making via hierarchical reinforcement learning," Sci. Rep. **14**, 4458 (2024).
3. A. Perrusquía, W. Guo, B. Fraser, *et al.*, "Uncovering drone intentions using control physics informed machine learning," Commun. Eng. **3**, 36 (2024).
4. M. Rosamilia, A. Balleri, A. De, *et al.*, "Radar detection performance prediction using measured UAVs RCS data," IEEE Trans. Aerosp. Electron. Syst. **59**, 3550–3565 (2023).
5. C. J. Li and H. Ling, "An investigation on the radar signatures of small consumer drones," IEEE Antennas Wirel. Propag. Lett. **16**, 649–652 (2017).
6. P. Janpangngern, P. Changnam, W. Charoensiri, *et al.*, "High-resolution FMCW radar for small UAV detection using GNU software-defined radio," IEEE Access **13**, 86396–86412 (2025).
7. L. Bi, Z.-X. Xu, and L. Yang, "Low-cost UAV detection via WiFi traffic analysis and machine learning," Sci. Rep. **13**, 20892 (2023).
8. I. Bisio, C. Garibotto, H. Haleem, *et al.*, "RF/WiFi-based UAV surveillance systems: a systematic literature review," Internet Things **26**, 101201 (2024).
9. M. H. Rahman, M. A. S. Sejan, M. A. Aziz, *et al.*, "A comprehensive survey of unmanned aerial vehicles detection and classification using machine learning approach: challenges, solutions, and future directions," Remote Sens. **16**, 879 (2024).
10. J. Zhu, J. Rong, W. Kou, *et al.*, "Accurate recognition of UAVs on multi-scenario perception with YOLOv9-CAG," Sci. Rep. **15**, 27755 (2025).
11. X. Yang, P. C. Konda, S. Xu, *et al.*, "Quantized Fourier ptychography with binary images from SPAD cameras," Photonics Res. **9**, 1958–1969 (2021).
12. R. H. Hadfield, J. Leach, F. Fleming, *et al.*, "Single-photon detection for long-range imaging and sensing," Optica **10**, 1124–1141 (2023).
13. A. McCarthy, G. G. Taylor, J. Garcia-Armenta, *et al.*, "High-resolution long-distance depth imaging LiDAR with ultra-low timing jitter superconducting nanowire single-photon detectors," Optica **12**, 168–177 (2025).
14. J. Rapp, J. Tachella, Y. Altmann, *et al.*, "Advances in single-photon lidar for autonomous vehicles: working principles, challenges, and recent advances," IEEE Signal Process. Mag. **37**(4), 62–71 (2020).
15. J. Rapp, Y. Ma, R. M. Dawson, *et al.*, "High-flux single-photon lidar," Optica **8**, 30–39 (2021).
16. Y. Hong, S. Liu, Z.-P. Li, *et al.*, "Airborne single-photon LiDAR towards a small-sized and low-power payload," Optica **11**, 612–618 (2024).
17. Z.-P. Li, J.-T. Ye, X. Huang, *et al.*, "Single-photon imaging over 200 km," Optica **8**, 344–349 (2021).

18. J. Tachella, Y. Altmann, N. Mellado, *et al.*, "Real-time 3D reconstruction from single-photon lidar data using plug-and-play point cloud denoisers," Nat. Commun. **10**, 4984 (2019).
19. X. Zhou, J. Sun, P. Jiang, *et al.*, "Improvement of detection probability and ranging performance of Gm-APD LiDAR with spatial correlation and adaptive adjustment of the aperture diameter," Opt. Lasers Eng. **138**, 106452 (2021).
20. A. Hou, Y. Hu, Y. Xie, *et al.*, "Full waveform recovery method of moving target for photon counting lidar," IEEE Trans. Geosci. Remote Sens. **62**, 5701114 (2024).
21. Z.-P. Li, X. Huang, Y. Cao, *et al.*, "Single-photon computational 3D imaging at 45 km," Photonics Res. **8**, 1532–1540 (2020).
22. Z. Xia, "Flash LiDAR single photon imaging over 50 km," Int. Arch. Photogramm. Remote Sens. Spatial Inf. Sci. **48**, 1601–1606 (2023).
23. A. McCarthy, R. J. Collins, N. J. Krichel, *et al.*, "Long-range time-of-flight scanning sensor based on high-speed time-correlated single-photon counting," Appl. Opt. **48**, 6241–6251 (2009).
24. Y. Zhang, Y. He, F. Yang, *et al.*, "Three-dimensional imaging lidar system based on high speed pseudorandom modulation and photon counting," Chin. Opt. Lett. **14**, 111101 (2016).
25. A. Maccarone, F. M. D. Rocca, A. McCarthy, *et al.*, "Three-dimensional imaging of stationary and moving targets in turbid underwater environments using a single-photon detector array," Opt. Express **27**, 28437–28456 (2019).
26. Y. Altmann, S. McLaughlin, and M. E. Davies, "Fast online 3D reconstruction of dynamic scenes from individual single-photon detection events," IEEE Trans. Image Process. **29**, 2666–2675 (2019).
27. T. Seets, A. Ingle, M. Laurenzis, *et al.*, "Motion adaptive deblurring with single-photon cameras," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* (2021), pp. 1945–1954.
28. D. Liu, J. Sun, W. Lu, *et al.*, "3D reconstruction of the dynamic scene with high-speed targets for Gm-APD LiDAR," Opt. Laser Technol. **161**, 109114 (2023).
29. H. Li, K. Zheng, R. Ge, *et al.*, "Noise-tolerant LiDAR approaching the standard quantum-limited precision," Light Sci. Appl. **14**, 138 (2025).
30. Y. Jiang, S. Karpf, and B. Jalali, "Time-stretch LiDAR as a spectrally scanned time-of-flight ranging camera," Nat. Photonics **14**, 14–18 (2020).
31. Y. Hu, A. Hou, Q. Ma, *et al.*, "Analytical formula to investigate the modulation of sloped targets using lidar waveform," IEEE Trans. Geosci. Remote Sens. **60**, 5700512 (2021).
32. D. G. Fouche, "Detection and false-alarm probabilities for laser radars that use Geiger-mode detectors," Appl. Opt. **42**, 5388–5398 (2003).
33. G. Shen, T. Zheng, Z. Li, *et al.*, "High-speed airborne single-photon LiDAR with GHz-gated single-photon detector at 1550 nm," Opt. Laser Technol. **141**, 107109 (2021).
34. Y. Altmann, X. Ren, A. McCarthy, *et al.*, "Lidar waveform-based analysis of depth images constructed using sparse single-photon data," IEEE Trans. Image Process. **25**, 1935–1946 (2016).
35. J. Tachella, Y. Altmann, M. Márquez, *et al.*, "Bayesian 3D reconstruction of subsampled multispectral single-photon lidar signals," IEEE Trans. Comput. Imaging **6**, 208–220 (2019).
36. A. Dudush, V. Tyutyunnik, I. Trofymov, *et al.*, "State of the art and problems of defeat of low, slow and small unmanned aerial vehicles," Adv. Mil. Technol. **13**, 157–171 (2018).
37. D. Liu, P. Xu, Y. Zhou, *et al.*, "Lidar remote sensing of seawater optical properties: experiment and Monte Carlo simulation," IEEE Trans. Geosci. Remote Sens. **57**, 9489–9498 (2019).
38. T. O. Hodson, "Root mean square error (RMSE) or mean absolute error (MAE): when to use them or not," Geosci. Model Dev. **15**, 5481–5487 (2022).
39. Z. Xu, K. Guan, N. Casler, *et al.*, "A 3D convolutional neural network method for land cover classification using LiDAR and multi-temporal Landsat imagery," ISPRS J. Photogram. Remote Sens. **144**, 423–434 (2018).
40. H. Ma, H. Ma, L. Zhang, *et al.*, "Extracting urban road footprints from airborne LiDAR point clouds with PointNet++ and two-step post-processing," Remote Sens. **14**, 789 (2022).
41. Y. Zhou and O. Tuzel, "VoxelNet: end-to-end learning for point cloud based 3D object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (2018), pp. 4490–4499.
42. A. H. Lang, S. Vora, H. Caesar, *et al.*, "PointPillars: fast encoders for object detection from point clouds," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition* (2019), pp. 12697–12705.